

TotalPlant Solutions (TPS) System

TPS Terms and Definitions

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About This Document

This document defines terms and definitions used in a TPS system.

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation, which if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation, which if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.

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or Symbols

.dll

Dynamically linked library. A library that is linked to applications programs when they are loaded or run, rather than as the final phase of compilation.

.exe

An executable file.

620 Logic Control System (620 LCS)

Honeywell's programmable logic controller. Serves as the base platform for both the S9000 and TPS Logic Manager.

or Symbols

620 Logic Control System (620 LCS)

A

Abnormal Condition

A condition other than normal that is detected while processing a sequence program in a Process Manager, Advanced Process Manager, or Multifunction Controller. A true condition causes the execution of an Abnormal Condition Handler.

Abnormal Condition Handler

A special routine invoked in a sequence program by Control Language statements and executed in the Process Manager, Advanced Process Manager, or Multifunction Controller, if system or process conditions other than normal are detected by the program. There are three types: HOLD, SHUTDOWN, and EMERGENCY.

Abstract

The information that defines a standard CRT screen display: what information it is to contain, where each item is to be located on the screen, etc.

AC Voltage Options

Require AC voltages for different devices. For example: 120, 220, 240 Vac +10%, -15%.

Access

To read or write a parameter value, or the content of a memory location

Access Level

For TPN:

Keyboard keylock position required in order to make changes to a given parameter. All data point parameters are assigned an access level that specifies what activities and tasks can change values in the parameter. TPN Security accommodates the following levels of access to the TPN Data Owners:

- OPER (Operator)
- SUP (Supervisor)
- ENGR (Engineer)

Access levels are fixed for some parameters, configurable for others.

Action Procedure

An action procedure is one or more actors that specify a desired action. Action procedures are also called **actors** (which overloads the term actor, defined below). Actors are triggered by touching a target or pressing on the configurable buttons on the Operator's keyboard. Action procedures are defined and associated with targets and buttons by users of the picture editor and button configurator.

Active Property

Active Properties are primarily used as triggers to invoke IOMap functionality when values are written to them.

Active Server

When using Redirection Manager, this term refers to the OPC server that is currently servicing OPC calls. Can be either the Primary or Secondary server. The Secondary server becomes the Active server when the Primary server fails.

Active Window

In the TPS Builder workspace, multiple project, library and drawing windows can be open at the same time. Only one window can be active at a time.

The active window is the one in which you are currently working. The active window is indicated by its title bar displaying the active window color (as defined by your system's display settings).

ActiveX

The de facto encapsulation technology for Windows

ActiveX Container

An application that supports the necessary Microsoft-defined COM binary interfaces (for example, IOleContainer) necessary to support the incorporation of ActiveX controls into a composite application.

ActiveX Control

A software component that supports the necessary Microsoft-defined COM binary interfaces (for example, IOleControl) necessary to support being incorporated into ActiveX container applications.

Actor

A pre-defined procedure that can be attached to a target on a custom display or to a button on the keyboard. When triggered, it causes some action to occur. For example, it could call up another display, change a mode, or display a port for entering data.

Add-in

A tool such as a wizard, a builder or a menu item that provides additional functionality to a Microsoft application

Address Assignment

The process of assigning points to the slots defined in a node. The Address Assignment process automatically assigns Input/Output (I/O) points to positions in I/O Modules and control points to positions in control processors. The number and types of I/O Modules present and the number of each type of control point present are specified when defining a node.

In TPS Builder, the user if required can change addresses assigned automatically.

An **I/O point** address is comprised of:

- Network Number
- Node Number
- Module
- Slot Number

A **control point** address is comprised of:

- Network Number
- Node Number
- Slot Number

Advanced Command & Control Operator Language (ACCOL)

Bristol Babcock's Advanced Communications and Control Oriented Language (ACCOL) is a symbolic language whose building blocks are ACCOL Modules and module signals.

Advanced Control Interface Data Point (ACIDP)

A data point whose database represents an Advanced Control Program in a Computing Module.

Advanced Control Program (ACP)

A program in a computing module that is represented by an Advanced Control Interface Data Point. Advanced Control Programs can be written in Fortran or Pascal.

Advanced Process Manager (APM)

A process-connected device on the UCN that provides regulatory control and sequence operations. The APM includes a powerful set of pre-built data acquisition and control algorithms, plus additional and enhanced data points as compared to a PM.

Alarm Summary Display

For GUS:

An NT-hosted display that is functionally equivalent to the Universal Station R520 Alarm Summary Display.

Alarm

Process variable behavior that results in an event

Alarm, Process

An indication (audible or visual) that notifies a console operator of an abnormal condition in the process

Alarms

Two types of alarm processing are performed by the CLM:

- System Alarms
- Point Alarms or Process Alarms

System alarms refer to the status of the network devices and the communications with these devices. The CLM Network Status display can be used to display the device and network status. Device failures are also output to the status notification journal. On R400 (and later) TPNs, device failures are also annunciated at the STATUS DETAIL level of the AM System Status display and by the System Status LED on the Operator Keyboard of the station. In this document, system alarms may also be referred to as communication alarms.

Point or process alarms refer to the status of the functions performed or monitored by the system. The status of these functions is maintained in the point database of the system.

Alert

For Fieldbus:

As defined by the Fieldbus Foundation, this is either an alarm, which has an accompanying return to normal, or an event that is simply the notification of an interesting condition.

Example: a device needs maintenance soon for which there is no return to normal.

Algorithm

A procedure for calculating a result, solving a problem, or accomplishing some ends. In **TotalPlant** Solution (TPS) systems, the algorithms do not actually reside in the data points that use them, but are retained in the memory of the process-connected boxes and modules that have them, and are used by the data points as the data points are processed; however, the algorithms operate as if they had continuous relationships with each other, and it is convenient to refer to them that way. For example, if algorithm A is said to be connected to algorithm B, you should understand that is true only as the data points using the algorithms are processed.

Alias

Alternative name for a given cartridge or HM volume in Release 200 and earlier releases. It has no meaning in R210 and later releases.

Alias Point

An HG (Hiway Gateway) data point that uses the same box and slot on a Data Hiway as another point in the same HG, but has a different tag name—an alias.

Alias points are created by building and loading only the HG-resident parameters for a point that already exists in the box and slot, and then loading the point in the HG. This new point in the HG then has access to the data in the designated box and slot. Alias points usually have virtually identical data, but different unit assignments. This allows alarms for such points to be generated for more than one unit.

Alias Variable

A variable that your application loads with an ASCII representation of an External Reference

Alternate Mark Inversion (AMI)

Bipolar data communication

Alternate Minimum Tax (AMT)

Part of '86 tax reform; incentive to lease

Alternate NG

The NG selected as the alternate communications path between the local system and a remote system connected to the same PIN. There must be at least two NGs at both the local and the remote system in order to establish responsible and alternate NGs.

Alternate NG Server

The alternate NG server is another name for the remote LCN's alternate Network Gateway. It is described in terms of its PIN local address.

Alternating Current (AC)

An electric current that reverses its direction at regular intervals.

Anchored Access

Anchored Access is the ability to specify the starting index of an External Reference array and read or write successive values into a HLL variable array.

Ancillary Variable

A general term used to refer to all EDB Data Access variables other than the Value Variable.

Annunciate

Displays LED lights and sounds horns attached to contacts. If an IKB is attached, the keyboard will beep. Otherwise, the PC will beep.

A

Anti-Reset Windup (ARW)

Anti-Reset Windup (ARW)

A function that prevents reset windup of PID points by inhibiting integral control action that is beyond the limits of the final control device (valve).

Anti-Windup

To limit the change in a control output to the maximum movement of the control device, so that the output will not "wind up" beyond the maximum movement of the device.

Application

A set of programmed or configured information files that are designed to be executed by a specific set of rules and/or environments.

An application is an executing program/process that has one or more user-visible displays, or **application windows**.

Application Builder and Executive (ABE)

Honeywell software package

Application Director

The TPS product package that provides an environment for the development and execution of advanced control applications. It consists of four components:

- An Execution Environment
- A Development Environment
- A set of standard displays
- A Step Processor

Application Module (AM)

A module on the Local Control Network that processes data points to provide advanced supervisory control. Both standard algorithms and custom programs written in Control Language can be used to process data points.

Application Module Personality (AMW)

AM Personality Image loaded into the LCNP portion of an APP.

Application Module X (AXM)

The AXM is a TDC 3000X advanced applications platform that combines the security necessary for advanced process control and optimization with the power to run complex applications from Honeywell and third-party suppliers. Existing Application Module (AM) functions - point processing, broad PID algorithm handling, and Control Language (CL) capabilities - are expanded in the Application Module X with the addition of a powerful RISC-based UNIX co-processor that lets you use a wide range of software applications from a variety of suppliers.

Application Processing Platform (APP)

The Application Processing Platform (APP) is the open application host for the Honeywell TPS system. The APP functionality is layered on top of the Windows NT operating system that makes it possible to integrate advanced control or information management applications.

Application Program Interface (API)

A set of protocols, routines and tools for building software applications.

Application CPU to Railbus CPU Interface Library (APRIL)

The APRIL software is executed by the APB's processor. The path for the APB's Application software to the Railbus CPU, and thus to the IOMs, is through APRIL.

Application Window

An application window is the display mechanism that allows an application to be visible to the user. An application window is usually one that is not a component of another (parent) window, but rather is an independently displayed window that has no user-interaction (for example, not a modal file open dialog box) with a parent window.

Area

A portion of the process, typically consisting of as many process units as one operator can handle competently and safely from one console.

Area Data Base (ADB)

The database loaded into all the Universal Stations that is a part of the console controlling a given process area. Contains definitions of:

- Logs and Reports
- Groups
- Alarming schemes for units and nodes
- Pathname for schematics

that are specific to a given area of control.

Arithmetic Unit (AU)

Number cruncher for a processor

Array Point

A type of data point that groups multiple data point types (flags, timers, numerics, and strings) under a single tag name.

Associated Document

An Associated Document is a document produced in a third-party application that provides information about a strategy. The association is made by:

- Embedding it in a drawing
- Linking it to a drawing, or
- Associating it to a strategy. Any OLE application, such as MS Excel or Word, can produce a document that can be associated with a strategy.

Attribute

Attribute is a parameter property. It determines how the parameter will “behave” in TPS Builder. Control attributes, include:

- setting parameters that will be used for connectors in a drawing
- which parameters will be listed on drawing symbols
- which parameters will be template defining, propagated down, or projected-up

Auto Recovery code for the RBP (ARR)

The primary purpose of ARR is to allow the RBP application code to be downloaded to the RBP. If ARR is present, it is entered on a controlled halt of the Railbus CPU.

Autoboot

A History Module that can reload and restart itself, if its power has been interrupted or if its power was turned off and then back on. This is referred to as an autoboot or autobooting.

AutoMan

Auto Manual Station Algorithm

Automation Solutions Center (ASC)

Honeywell group that demonstrates products for customers at Honeywell site or at various trade shows. Previously known as: ATC and GSC.

Auxiliary Port

An RS485 port on the IOLBIM’s RBP primarily intended for configuring and monitoring HART devices. This port will also be used for several other “Auxiliary” purposes by connecting the Technician Utility (see below).

Availability

The proportion of the time a system (or subset) performs to certain criteria (specifications) relative to the total time.

Average RMS Current

As explained in the **TPS System Site Planning** manual, most current meters simply rectify and average the AC current and have a scale that is calibrated to indicate the RMS value, assuming a sinusoidal input.

B

Backbuild

In general terms, backbuilding is the process of transferring control system data away from the control system, usually into an off-line database, for the purposes of reconfiguration or documentation.

Within TPS Builder, backbuilding is the importing of TPS system entity data (point and node), from Exception Build (EB) or other files, into the TPS Builder database. These entities appear in the form of TPS Builder strategy blocks (and in the case of Logic and Device Control points, strategy modules), within a selected project.

Backplane

A printed circuit board, located in the back of the chassis, providing data bus, power bus, and mating connectors for all installed modules.

Backup

Hardware - a device capable of taking over the functions of another device.

Software - a copy of some set of data on another disk drive, local or remote.

Backup Domain Controller (BDC)

The Windows NT Server Backup Domain Controller (BDC) maintains a copy of the Security Accounts Manager (SAM) database. BDC's provide fault tolerance and load balancing for client authentication.

Bad Control (BC)

For AM points. This is an indicator in the AM data points that use control algorithms. It is accessible to programs and to other data points, and may appear on Universal Station displays. When set, it indicates that something has happened that prevents the data point from properly processing the control algorithm.

This is an indicator similar to the Bad Control indicator that, when **Return** is set, indicates that the Bad Control indicator has been reset and proper control is now possible.

B

Bad Value Status (BP)

Bad Value Status (BP)

The value of the PV (Process Variable) is indeterminate. This might be caused by an unavailable input, a PV range violation with no clamping, an invalid PV constant, or a bad input-value status. For example: Bad PV=BP.

Bar Chart

A rectangular display object whose height (or width) can be controlled by using an **expression** in a picture.

BasicScript

A Basic Scripting Language by SUMMIT, similar to Microsoft Visual Basic. This is the foundation for GUS Scripting.

Batch

A result achieved or a product created by one or more sequence-program executions.

Batch Control Strategy

The control plan used to achieve an end result. The result can be an intermediate product or a product made from a complete recipe.

Behavior

A particular characteristic attached to the display of a graphic object: color, color intensity, blink, and video status (normal or reverse).

Bernoulli

A cartridge disk or cartridge disk drive; the cartridge has a storage capacity of approximately 20 MB (megabytes).

Bias

Offset applied to a controller input

Binary

Compounded or consisting of or marked by two things or parts.

Bind

For Fieldbus:

Designates any operation that takes an object or data reference from a general form into a more specific form. For example, connectable parameters bind to a specific data source.

For TPS:

The conversion of external entity and parameter names to the internal entity ID and parameter ID, as used for LCN communications. The external/internal conversion can be performed by only the owning node of the external entity. This means the entity to be bound must be built and loaded, and the owning node must be running on the LCN, at the binding time.

Binding

The process of replacing virtual references with actual references; also called linking.

Bit

Binary 1 or 0

Blend Property Control (BPC)

Refinery software package

Blend Ratio Control (BRC)

Refinery software package

Blind Record Data Pipe (BRDP)

The BRDP is an information “tunnel” through TPS networks from an origin on a TPN Server client.

Block

The lowest level element that makes up a template or strategy is a block.

- In the TPS Process Network, blocks equate to points.
- In the History Module (HM), 1 block = 1 Sector = 256 Bytes.

In addition to the engineering aspects of a TPS Process Network point (such as parameters, behavior, or programs if applicable), a block also carries with it TPS Builder-specific characteristics, such as TPS Builder property information and how it is represented in a drawing.

Boiler Turbine Generator (BTG)

Power Generation control panel

Bootload

The loading of a program or data into a node or device directly without intervention through some other node or device.

In the **TotalPlant** Solution (TPS) system, loading a Universal Station by itself and not through the Console Status display at another US is referred to as bootloading the US. Also, a History Module can bootload itself (also see Autoboot).

Bose Chaudhuri Hacquenhen (BCH)

Data transmission error checking technique

Bound Data Point

A program in CL (Control Language) can execute only after it has been connected to a data point by the CL Linker. This data point is known as the program's Bound Data Point. The Bound Data Point of a Sequence Program is a Process Module Data Point; a CL Block is bound to a Continuous Data Point.

Box

A process-connected device that connects to a Data Hiway. Refers to a Hiway Box such as Basic, Extended, Multifunction, or Universal controllers.

British Thermal Unit (BTU)

The British Thermal Unit is equivalent to the quantity of heat necessary to raise one pound of water one degree Fahrenheit (at 39°F). It provides a measurement of power converted to a form that is useful for sizing heating and cooling systems. The conversion from Watts to BTU/hr is:

$$\text{BTU/hr} = 3.41 \times \text{Watts}$$

Browse

Screen-based method of locating items in a control system

Buffered, Network-Initiated, Unidirectional (BNU)

For Fieldbus:

A type of VCR (Publish, Subscribe)

Build

Entities are built by interacting with the Data Entity Builder's Parameter Entry Displays (PEDs) or by exception building. The entities can then be stored into an IDF for future loading to the owning node. Building *does not* include loading in the owning node.

To enter configuration data, picture information, reports, and logs into the system through one of the facilities of the Engineering Personality, such as the Data Entity Builder.

Build Mode

One of two standard modes in which ActiveX controls operate. Build mode indicates that the ActiveX control is not “running” or “operational” (see *Run mode* definition). Rather it behaves as appropriate for configuration/build/design activities.

Builder

Engineering tool for creation of items in a control system

Build Type

A category of points based on a master list provided in the point builder (Data Entity Builder) of the Engineering software. Examples: regulatory points, digital points, process module points.

Builder Input File (BIF)

The BIF may contain all of the CLM interface structure definition entries required to establish the CLM interface between the LCN and the non-TPN network. In its completed form, the BIF represents source input to the CLM Configuration Builder, using the file input mode.

Bulk Build

Bulk building is a TPS Builder function that uses Microsoft Excel to efficiently create new strategies that are similar. The more you know about Excel, the more efficiently you will be able to create multiple strategies using this function.

Bureau of Labor Statistics (BLS)

Price escalation factor

Button Service

A Honeywell-provided software component provided on GUS nodes and other non-desktop configurations, for the purpose of directing IKB keystrokes to the appropriate application.

Byte

A group of eight data bits (binary digits) operated on as a unit.

One word equals 2 bytes in the TPN.C

B
Byte

C

Cabinet (CAB) File

A single compressed file format used to store multiple files. A CAB file contains a compressed version of the file(s) needed by an ActiveX control plus a file with instructions for the Internet Explorer, specifying how the control is to be installed.

Cable Fault

A communications malfunction that causes a complete disruption of communication, an excessive error rate, or unrecoverable errors. Use the LCN Diagnostic (\$LNMENU) display to troubleshoot LCN cables, or the UCN Statistics and other displays for UCN cables. Also known Suspect Cable

Cache

Whenever OpenDDA accesses a DCS to resolve an External Reference, the information returned (Internal Id, data type, etc.) is entered into a Cache located in the HP-UX. Subsequent resolutions for this External Reference are satisfied by the Cache instead of accessing the DCS, which greatly improves performance.

Cache Management Tool (CMT)

The Cache Management Tool allows examination and/or deletion of Cache entries. It also supports checkpointing the cache.

Canadian Electrical Code (CEC)

Canadian standard for electrical installations

C

Capabilities File (.cff)

Capabilities File (.cff)

A device manufacturer supplies a Capabilities File. The Capabilities File is a text file that is written in FF Common File Format.

The Capabilities File describes the communication profile, device capacity, internal record address assignments, and certain default values for a field device.

Capability Maturity Model (CMM)

For software SEI

Cartridge

A type of magnetic storage disk or disk drive; the cartridge disk used on TPN systems has a data storage capacity of approximately 20 MB (megabytes).

Cascade

Control scheme in which the set point of a loop is the output of another loop

Category

See Window Category

Cathode Ray Tube (CRT)

A charged-storage tube where information is written using a cathode ray beam found in monitors and televisions. CRT is a device that electronically displays data or graphics on a screen for viewing. Although CRT refers to the actual tube itself, the device is sometimes also called a VDT (Video Display Terminal) or simply a monitor.

Central Processor Unit (CPU)

The unit of a computer system that includes the circuits controlling the interpretation of instructions and their execution.

Central Segment

Central coax segment to which LCN Extenders can be connected to extend the LCN system.

Centralized Operation

Use of an operating center to control the operation of widely distributed process units and distributed processing equipment.

CFF File

See Common File Format.

Change of State (COS)

Alarm detected when the state of a digital input point changes in either direction.

Channel

Generally used for a single input or output on an IOM. Same as a PM family's Traditional IOP's Slot.

Characters Per Second (CPS)

Printer speed that cannot be configured.

Chassis

The card cage where the boards for a node reside.

Checkpoint

Storing of the process database from a module or gateway in files from which the data can be reloaded into the module or gateway, should it be necessary to restart it. Where a gateway serves process devices, such as the boxes on a Data Hiway, their process data can be included in the checkpoint files, and reloaded. The term checkpoint arises from the notion that a snapshot of the data is acquired at some point in time—a checkpoint.

Manual or automatic periodic storage (saving) of operating data onto a nonvolatile storage medium, such as the History Module or a removable medium, to provide a backup database for cases of extreme failure in a device's memory.

CL Block

The unit of CL (Control Language) execution in continuous control. One or more CL Blocks can be attached to insertion points of any continuous data point. A CL Block can also be used in place of the PV Algorithm or Control Algorithm of a regulatory data point.

CL Editor

A tool provided with TPS Builder that allows you to efficiently write CL programs using Premia's Codewriter (version 5.0c), to which CL functionality has been added. In addition to all of Premia's features, CL Editor provides

- Complete CL editing
- CL syntax checking, and checking against the TPS Builder database
- Online CL error help

A TPS Strategy Browser for locating TPS Builder strategy parameters

CL-INIT Application

An application built to run on an amw and is initiated by a CL Server. It is a window-less console application.

CL Switch

A Control Language (CL) feature that allows changes in control strategies through the use of CL Blocks. It is implemented with a collection of parameters added to a Regulatory Data Point or by using a CL Switch Data Point. An operator at the Universal Station or a program can direct the switching strategies.

Client

A user software application or computer that requests the services, data, or processing of another application or computer (the server). In a two-task environment, the client is the user process. In a network environment, the client is the local user process and the server may be local or remote.

Clipboard

A temporary storage space where you can move text (or images) from one application to another, using the Copy, Cut commands. For example, you can use the clipboard to copy text from a word processor document and insert (paste) it into a dialog text box in another application..

CLM Image Point

The CLM image point is a TPN based data point (usually an AM type) that is used to emulate the non-TPN point or data address. This point is the basic mechanism that makes the non-TPN network data point appear to be directly connected to the TPN. The CLM image point emulates its corresponding non-TPN network data point, through the CLM software and point interface structure definitions. The CLM image point is created through the Data Entity Builder in the TPN Engineering Personality.

Clock Source Repeater (CS/R)

I/O size board that transmits a 20 ma current loop signal to keep LCN nodes synchronized to the same timestamp in a mixed or non-KxLCN processor system.

Also acts as the coax cable ground.

Cluster

Defined set of TDC 3000X nodes used as basis for loading, performance and cost estimates

Coating and Graphic Arts (C&GA)

Division of TAPPI

Coaxial Segment

A continuous connection of LCN nodes by a dual redundant pair of coaxial cables. A coaxial segment is limited to 300 meters maximum and can have a maximum of forty loads including LCN Extenders.

Coexistence

Co-resident

Characteristic of two or more programs running in the same processor/module or node

Collector

A collector is a Honeywell-defined function that accesses data other than process and display database data. A collector can be used in **expressions**. Collectors cannot be called from actors (a.k.a. action procedures). Collectors can use the data scanning behavior of the data access subsystem to periodically fetch data and display the data in schematics.

Comite Consultatif International Telephonique et Telegraphique (CCITT)

Part of International Telecommunications Union

Comma-Separated Values (CSV)

Comma-Separated Values as in .csv file formats

Common File Format (CFF)

The Common File Format is a Foundation Fieldbus specification. The common file is a text document formatted like a Windows INI file.

The file contains some or all of the information about a single device that can be read from a fieldbus device over the wire. There is two substantially different type of common files, capabilities type and value type.

Communications Line Interface (CLI)

An interface board used in Computer Gateways that is compatible with a wide variety of devices. It has an RS232 serial interface and an IEEE 488 parallel interface.

Communication Link Module (CLM)

A TPN Application Module (AM or AxM) with at least one CLM Interface board and the CLM software package installed.

CLM is a module that resides on the TPN and provides a smart interface to certain third party devices such as tank gauging instruments and specific programmable controllers.

The CLM requires a minimum of 6-megaword memory, and TPN Release 4xx (or later) revision. It is used to communicate to different third party PLC devices.

Communication Master

A communication master is a system, which is able to initiate an exchange of data with other systems without being requested by another system.

Communication Server

A communication server is a system, which is not an end user of the exchanged data, but serves as a routing device between master and slave systems.

Communication Slave

A communication slave is a system, which only exchanges data with another system only when requested by the other system.

Communications Link Module TIWAY (CLM-T)

Texas Instruments tank gauging system

Compel Data (CD)

The data link protocol unit used by Function Blocks that publishes data on the Fieldbus Link.

Compile

To convert a high-level computer language, such as Control Language or Fortran, from the human-oriented form in which it is written (called the source code) to the machine-oriented form that a microprocessor can use (called the object code).

For TPN:

Custom (schematic or graphic) displays, CL programs, free format logs, and button configurations are compiled from source data (either stored, edited images, or textual source programs) into final object code that is actually used at runtime, by the appropriate function. In general, direct references to entities in the system must be bound at compile time. This means the entity must be built and loaded, and the owning node must be running on the LCN, at compile time.

Component

A piece or part of something. Used in context, it may be taken to have any of a number of specific meanings, such as:

TPS Component A software entity that is part of the TPS product.

HCI Server Component An HCI Server which minimally supports OPC interfaces. An HCI server component may be in-process or out-of-process.

HCI Managed Component A TPS component, which runs in its own process and which supports the COM interfaces necessary to be managed by TPS System Management. Current HCI Managed Components

In other contexts, “component” may refer to the pieces of an application, or pieces of hardware.

Component Administration Service Provider

CAS Provider represents a WMI provider that sends event notifications to WMI clients when status of OPC servers change.

Component HCI

An HCI component. They are listed in the tree display, under Nodes on the TPS System Status Display. There are both Managed and Client Launched Components.

Component (Client Launched)

An HCI component is a component that is launched by a client. When the client releases the component, it triggers the component shutdown.

Component (Managed)

An HCI component that can be started by TPSAdmin (AutoStart) or manually from the TPS System Status Display.

Component Object Model (COM)

Component Object Model is the Microsoft standard (the COM Specification) for component architecture. The purpose of this specification is to ensure that programming components and clients can operate together.

COM is a component software architecture that allows applications and systems to be built from components supplied by different software vendors. High-level OLE application services are built on the common Component Object Model foundation.

Composite Point

A data point that combines more than one process input or output in a single data point. A composite point can combine points only within the scope of a given device. For example, Digital Composite Point.

Composition Control Relationship

The relationship between a strategy module and a strategy it contains, or a template module and the template it contains.

Composition View

The Composition View is the hierarchical organization of strategies in a project. This organization is defined by the user, but does follow some basic rules. A user can define different project views, such as:

- Hardware – a hierarchy of network, node, and points.
- Equipment – a hierarchy of area, equipment unit, module, and points.
- Control – a hierarchy of control aspects, kept separate from the equipment type of strategies)

Users can also create their own views.

Computer Gateway (CG)

An LCN node used to interface to other computers such as: Digital's VAX and IBMs.

Computer Interface Unit (CIU)

Applies only to releases 135 and earlier to interface the LCN to other computers.

Computing Module

A computer system connected to the Local Control Network, consisting of an upper level host processor plus a Plant Network Module or Computer Gateway that provides user-developed high-level control and plant-wide information integration.

Computing Module 50N (CM50N)

Interface between the TPN network and VAX

Condition Handler

See Abnormal Condition Handler.

Conditional Behavior

Conditional behavior can be defined for picture objects that determine the object's color, intensity, and blink state based on process data by using a simple textual language. Conditional behavior is defined by using the ADD BEHAVIOR command in the picture editor. The conditional language combines if statements with the **expression** syntax to define the conditions.

Conduit

A pipe, tube, tray, or pan for protecting electrical cables and wires.

Configuration

1. The definition of the process database and the content of each data point
2. A specific arrangement of system equipment to meet user needs
3. Development of a human interface display structure to match the plant operating design

Configuration Device

For Fieldbus:

Network-connected device configuration tool

Configure

For hardware, this means to select the mix and quantity of hardware devices needed to meet the user's needs.

To configure also means to specify the information the data acquisition and control subsystem needs to cooperate in the management of a specific process or plant. Configuration can include:

- Filling out configuration forms and doing other paperwork
- Entering configuration data for points and programs through the Universal Station's Engineering Personality.

Conformite (or Communaute) Europeene (CE)

Mark indicating EC compliance for European country specifications.

Connection

In a control drawing, connection is a line connecting two drawing symbols, indicating input or output from one to the other.

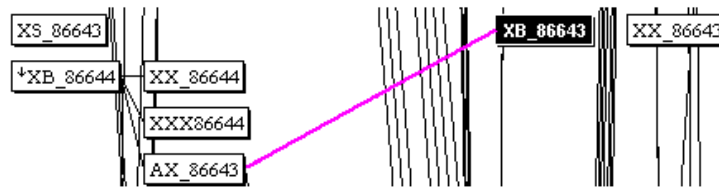
In cabling, connection is the communication path from node to node, or from main segment to remote segment using fiber optics and LCN Extender hardware.

Connection Manager

The connection manager manages connection information about device types, devices, and blocks that are actively communicating with a host application. The DD Viewer implementation of this service is an offline implementation.

Connectivity Map

A connectivity map displays TPS system entity strategies as a network diagram of strategy names and interconnecting lines that represent configured connections. A connectivity map can be generated for all selected Process Managers (PM) family entity strategies within a particular project.



Connector

In a control drawing, a connector is a short line protruding from one side of a drawing symbol. It represents the data flow input or output.

Connectors are also distinguished as passive and active. Passive connectors are lines alone, active connectors include an arrowhead.

Console

From one to ten Stations (US, UxS, or GUS) configured to function as a console. A console is typically used by one process operator.

Constant Expression

An expression that evaluates to a number or string constant. These exist to simplify the use of common data, such as the size of a particular display region.

Contained Parameter

For Fieldbus:

A block parameter which is accessible to the communication network but may not be linked to an input or output parameter. The parameter value may be used in the block algorithm or written by the block algorithm.

Context Records

For Application Director:

The configuration files for an application and its components (Sequence and Steps) such that, when installed into an Execution Environment, the application can be recognized, controlled and managed. Both static and runtime data are contained in the records.

Continuous Data Point

A data point whose processing functions are the same at each processing cycle. Regulatory, timer, and custom data points are examples of continuous data points.

Continuous Process

A process where raw materials are continuously processed and product is continuously drawn off, such as petroleum refining. A process that is not frequently (daily or hourly) started up and shut down.

Control Attributes

A control attribute refers broadly to two aspects of configuration: drawing symbols, and ways to improve configuration efficiency (and consistency). The function applies to templates and strategies (both blocks and modules). The attributes differ depending on what item is being configured.

Drawing Symbol Appearance: Determine the presentation of connectors and parameter information on drawing symbols

Configuration Assistance: Set template defining parameters, propagate-down parameters, and project-up parameters.

Control Drawing

A Control Drawing is a graphical representation of strategy or template modules and their connections. It may include associated or attached documents, and associated GUS pictures. A strategy / template module may have multiple control drawings showing the different aspects of the control.

Control Language (CL)

The process engineer's programming language. A subset of this language (CL/C) is applicable to continuous data points. Another subset of the full language (CL/MC) is used to create sequences that execute in the Multifunction Controller.

CL (Control Language) is a Honeywell proprietary software language used to build programs, which will execute in an Application Module (AM).

Control Language for the Multifunction Controller (CL/MC)

CL/MC is functionally similar to SOPL. See Control Language.

Control Module (CM)

For Fieldbus:

A new TPS point type. The Control Module point type has three subtypes: the Segment Entity Control Module (SECM), the Physical Entity Control Module (PECM), and Function Block Control Module (FBCM).

Control Output

The output of a controller. It may be an analog output signal to a process control device, it may be a digital data word from one location to another in the same device or from one digital device to another, or it may be the output of a contact set.

Control Panel

The standard human interface dialog box that is available with the GUS SafeView product. This control panel, which can be shown or hidden using the Ctrl-Alt-W keystrokes at any time, provides the operator or engineer the means to control SafeView.

Control Processor

A component of the Process Manager Module in the Process Manager, and Advanced Process Manager Module in the Advanced Process Manager, that performs all control processing. The Control Processor processes those types of data points that are not processed by the I/O Processors.

Control View

A project view that shows a hierarchy of control aspects, kept separate from the equipment/hardware type of strategies. The focus is on the more advanced control aspects of a plant, for example, optimization.

Controller Box (CB)

TDC 2000 Basic Controller

Conventional Kraft (CK)

Pulp digester

Conversion

Projects created in Honeywell Point Builder or LCN Workbook can be converted to TPS Builder projects. The following project elements are converted:

LCN Workbook project: TPS Network points and node configurations

Point Builder project: TPS Network points and node configurations, user-created templates (Drawings are not converted.)

Crash

The unexpected failure of a node that is caused by a situation the software or firmware in that node cannot manage or under which it cannot continue to operate. For example, as a node is loaded, the software checks for proper hardware revision levels, and if they are not proper, loading stops and the node crashes. The status indication on displays is FAIL.

Critical Process Controller

A name for the Triconex Fault Tolerant Controller that connects to the Data Hiway through a Hiway Interface Module.

Cross-screen

Ability to affect what is happening at one screen from another

Current WebGUS Environment

The WebGUS environment definition currently **active** on the user Desktop. A user may have one and only one **active** WebGUS environment.

A Desktop user selects a WebGUS environment from a list of environment definitions and sets it as the **Current Environment**. The action of setting the selection as the **Current Environment** applies its operational settings to the Desktop's Registry.

Cursor

An indicator on a CRT screen that can be moved to identify a location for selection or entry by the user.

Custom Data Point (CDP)

A continuous data point consisting of functions performed by CL Blocks, using Custom Data Segments.

Custom Data Segment (CDS)

Data whose characteristics are defined to the system by the user and is incorporated into data points. Part of an Application Module data point used for storing data that is to be used by CL programs.

For example, the name of each parameter and its type, units, and change authority are included when the segment is defined. Once defined and attached to specific data points, the CDS is automatically available for displays and other functions.

Custom Data Segment Description

The user-defined descriptions of the parameters of a Custom Data Segment. This definition is implemented through the CL Compiler.

Custom Display

See Schematic.

Custom Parameters

User-defined parameters added to the standard parameter list of a template. The only TPS Network-loadable custom parameters are in the AM as custom data segments (CDSs).

Other custom parameters can be maintained only in the TPS Builder database, usually for supporting engineering activity, for example, memory unit (MU) usage, processing unit (PU) usage, and estimated engineering effort.

D

Data Access

The set of routines that provides access to the various databases in the system.

Data Access Protocol (DAP)

Similar to DECnet

Data Access Set

A Data Access Set is a collection of EDB statements whose Value Variables and Ancillary Variables will be acted on in one operation such as an EXEC DDA READ or EXEC DDA WRITE statement.

Data Entity

A named collection of data; for example, a data point.

Data Entity Builder (DEB)

A function accessed from the Universal Station's Engineering Main Menu that is used in building data points. Building includes accepting configuration data, verifying that data, storing it in intermediate data files, and loading it in the destination module or gateway.

The DEB operates independently from the type of entity that is being built. The Global Description Files (GDFs) allow it to do this. The GDFs provide the screen forms for the Parameter Entry Displays; they define to the DEB what data is to be entered and built, and how the syntax and definition are to be checked.

Data Hiway

Honeywell's proprietary, coaxial-based, dual communications network that interconnects specified Honeywell process-connected devices with one or more Hiway Gateways.

See Hiway

Data Hiway Port

A Data Hiway box that allows TPN equipment to communicate with Programmable Logic Controllers (PLCs) and other devices adhering to the same protocol.

Data Link

Communication channel over which data is transmitted between devices.

Data Measurement Corporation (DMC)

Measurex Industrial Systems Division

Data Owner

The software function that is the caretaker of a part of the system database, such as a data point. Data owners are responsible for validation of requests for the data they own, and for validating attempts by other functions to change the data they own. For example: AM, CG, HG, NIM.

Data Point

A TPN data point is a named collection of related parameters and processing involved with the acquisition or calculation of a main item of information and may optionally include the data for an associated control loop. A data point has three parts: a tagname (also called point name or point ID), parameters, and functions.

A named collection of data values called parameters. The data point forms the basis for a set of functions that operate on the collection of values.

Data Point Attribute

The characteristic of a data point that determines whether the operator or a program controls the setpoint or output.

Data Point Mode

The characteristic of a data point that determines how the data point operates; for example, automatic mode for a Regulatory Control Point specifies that the point's setpoint is to come from the Universal Station, while the point calculates its output according to its configured algorithm. Manual mode, on the other hand, instructs the point not to execute its algorithm and to accept an output value directly from an external source, such as the Universal Station or a program in a module or process-connected device.

Data Point Parameter

One of the pieces of data in a data point, consisting of a value (for example, real, integer, logical, and string), or an array of values. Data point parameter examples are setpoint, process variable, and high alarm limit.

Data Point Residency

The location (module or process-connected device) where the data point exists.

Data Point Type

See Build Type.

Data Source

A type of target system where data is either manipulated using a non-procedural language, such as SQL, or where the data manipulation can be mapped in a non-procedural manner.

Data-Flow Connection

A relationship between two strategies / templates which represents the flow of data from one to the other, for example, FIC1.OP to FIC2.SP.

Database

A set of files, treated as a unit, in which a server stores a set of data dictionary tables and user tables.

For TPN:

All of the data stored in all TPN modules and process-connected devices. The database contains the information that the software and firmware programs use to execute their functions.

Database Client

A client, usually an application or utility, running on a PC or other computer, that requests the services of a database on a remote computer (host).

Database Server

The node or computer at which the server kernel runs. The database server holds the database.

DCOMCNFG

Microsoft-provided tool for configuring DCOM settings

DD Binary

DD (Device Description) is a binary file that provides the definition for parameters in the FBAP (Function Block Application Process) of a device.

DD Item

A fundamental concept of the Device Description Language is the **item**. An item is a major component, which makes up the description of the device and can be any of the following constructs:

- **Array**
- **Block**
- Collection
- Domain
- Edit Display
- Item Array
- Menu
- Method
- Program
- **Record**
- Refresh Relation
- Response Code
- **Variable**
- **Variable List**
- WAO Relation

Note: The DD Viewer, as a template-building tool, is mostly concerned with the **bold** DD Items above.

DD Revision

To determine the latest version of the device type, Fieldbus Foundation incorporated the Device Revision and the DD (Device Description) Revision into the name for the DD binary and symbol file.

The left hexadecimal byte of the name represents the Device Revision.

The right hexadecimal byte of the name represents the DD Revision.

DD Synthesizer

For Fieldbus:

A FF (Fieldbus Foundation) supplied tool that combines incremental DD's, with unresolved references, with Fieldbus Foundation Standard DD's to produce a complete, or full, DD that can be used with DD Services.

DDO file

The file extension given to incremental DD binary files, from a device manufacturer, that is to be converted to full and complete DD binary files by the FF Synthesizer.

Debounced Data

The data of the multiplexer is read if the data on the data bus is stable. The data on the data bus is stable if two consecutive samples on the input data bus of the multiplexer return the same data result.

Decibel (dB)

Decibels are the measurement of noise or sound, which follows a logarithmic scale. The threshold of pain is 120 dB. Normal human hearing starts at 0 dB.

Declared Data Point

A data point that is not bound to a particular Control Language program but is accessed by the program to obtain input for a process calculation.

Default

A default value is a value that the system uses if the process engineer elects not to configure a parameter. Default values are built into the system to eliminate the need for the engineer to enter values for parameters that are not used, or to let the system use a typical value.

Default Match Expression

A match expression used to manage registered application displays that do not match any of the current workspace's match expressions. Allows displays with new category values, not known to the current workspace configuration, to be managed in this workspace. The DEFAULT keyword is used to for this purpose.

Default NWDDDB Database

An Input File of OPC data items with default characteristics. The file is loaded into RAM at NWDDDB Server start-up. It has the well-known name "nwddb_server.dss".

Default Value

A pre-chosen value for some configurable parameter that is used if no other value is substituted for it.

Default WebGUS Environment

The WebGUS environment definition established as the initial **Current Environment**, when the WebGUS control is first downloaded to the Desktop.

A Site Server administrative user creates a set of environment definitions and designates one as a **Default Environment**. The **Default Environment** becomes the **Current Environment** transparently, when the WebGUS control downloads to the Desktop.

Depopulate HLL

High Level Language variables associated with the External Data Statement will be moved to OpenDDA in preparation for writing to an external location.

Desktop

Equipment targeted at other than Operating personnel in a plant. Also equipment used by operators who do not have consoles.

Deterministic

For Fieldbus:

A signal, the future of which can be determined precisely.

Development Environment

For Application Director:

The environment which an application and its components are configured and developed to run in Application Director.

Device Description (DD)

Field Device Object Dictionary descriptions of device applications can be supplemented by Device Descriptions (DD) generated from the Device Description Language (DDL).

D

Device Description Interface (DDI)

Device Description Interface (DDI)

DDI is the API exposed to the application so that the DD Binary can be loaded and read for device descriptions.

DDI provides the following kinds of interface functions:

- A function for getting DD items
- Convenience functions for getting auxiliary information about a DD item.
- Language functions for processing DDL strings
- Directory table functions to read device and block directories

Fetch and Eval are lower level functions called by DDI to extract and evaluate data contained within the DD binary.

Device Description Language (DDL)

The language that a device developer uses to define their FBAP (Function Block Application Process) blocks and parameters.

Device Description Services library (DDS)

A software library developed by Foundation Fieldbus that provides a generic access to a DD (Device Description).

Device Macrocycle

For Fieldbus:

A time interval that represents a single iteration of the function block schedule within a device. A block may run more than once in the macrocycle if it is entered more than once in the function block schedule.

Device Revision

To determine the latest version of the device type, Fieldbus Foundation incorporated the Device Revision and the DD Revision into the name for the DD binary and symbol file.

The left hexadecimal byte of the name represents the Device Revision.

The right hexadecimal byte of the name represents the DD Revision.

Diagnostic Program

A program that tests system equipment and diagnoses a failed or defective Optimal Replaceable Unit.

Digital Certificate

A certificate issued by a certificate authority, like VeriSign, which identifies an ActiveX control as trusted. The certificate data set indicates the name and digital ID of the control's creator.

Digital Enhanced Communications (DE)

Smart Transmitter Protocol

Digital Equipment Corporation Network (DECnet)

A networking protocol developed by the Digital Corporation that allows users to access information on remote computers.

Dimethylsulfide (DMS)

Pulp reactor chemical

Directory

A named subdivision of a History Module (HM)/cartridge volume to which one or more files is assigned.

History Modules/cartridge can have several volumes.

System-reserved directory/volume names begin with "&" or "!" and can be up to four alphanumeric characters. For example: DIR3

Disable

For Point/Unit alarms

A command that disables reporting of point alarms to the Universal Stations, on a point/unit basis, respectively.

Disable can also be used for other purposes such as history collection and checkpointing.

Discontinuous Process

A process operation that is frequently started, stopped, or held in a steady state.

Display Builder

The Honeywell tool for building graphical, dynamic operator displays representing process data.

Display Building

The creation of custom displays, such as graphics.

Display Database (DDB)

The Display Database (DDB) is scratch pad memory available to pictures for use in storing application-defined data. The DDB is accessed by using the actor language and expressions. There are two DDBs, the **Global** and **Local DDB**. Data stored in the global DDB persists between picture invocations and data in the local DDB does not.

Display History Dialog

A dialog box containing a list of up to 50 of the most recently displayed (and removed) applications. This dialog can be used to invoke a formerly displayed application.

Displays

See Schematic and Standard Displays

Disposable

Control outputs from regulatory data points are sometimes referred to as disposable or **indisposable**. To be disposable, the output must be connected to an input of at least one active secondary data point in CAscade mode, through a configured, active input; otherwise, the output is indisposable. Indisposable is a situation similar to the Initialization Manual condition in controllers (CB, EC, and MC) on the Data Hiway.

Distributed Control

1. Physical distribution of control for reliability and installation cost savings
2. Functional distribution for system testing and system performance improvement.

Distributed Control System (DCS)

Computer systems that are at different locations throughout a plant used to control the process. For example, TPS (TotalPlant System)

Distributed Process Controller (DPC)

DPC 3330 A - DPC package for mounting on a wall, or in a NEMA 4 enclosure

DPC 3335 A - DPC package for rack mounting

Domain

A domain is a logical grouping of Windows NT network servers, other computers, and devices that share common security and user-account information. It can be administered as a single unit with common rules and procedures.

Domain Component

A component that can be accessed by any client within a domain and its alias name is unique within the (OU) TPS Domain. The alias name of a Domain Component cannot be the alias name of a local-only component.

Domain Controller

A domain controller is a computer running Windows NT Server that uses one shared accounts database to store security and user-account information for the entire domain. A group of domain controllers comprises a single administrative unit.

Double Cable Fault

A cable fault on both cable A and B at the same time. This is a catastrophic incident in which redundant node types (NIMs, HGs) communications fail because of a cable fault of the LCN cables.

Download

Download is the transmission of CM point, Fieldbus link, and Fieldbus device data from TPS Builder to the TPS Network with Fieldbus.

Download is also used to send data to smart transmitters.

Drag

A standard mouse technique for moving objects. Refer to Windows NT Help for information on standard techniques. Choose Start > Help.

In TPS Builder, dragging sometimes causes an object to be copied, and sometimes to be moved. For example, dragging a template to a project or drawing will cause the template to be copied (to create a strategy), but dragging a template from one library to another will cause the template to be moved.

Drawing Information

Identification information that appears on the bottom of each page of a printed control drawing:

- Project name
- Drawing title
- Version
- Engineer
- Drawing no.
- Notes
- Page number

The drawing author enters this information in conjunction with a print request.

Drawing Symbol

A Drawing Symbol is the graphical representation of a template or strategy block or module in a control drawing.

Duetsche Institute fur Normung (DIN)

Standards organization

Dun and Bradstreet (D&B)

Company credit ratings

Dynamic Data Exchange (DDE)

Microsoft interface used in GUS displays.

Dynamic Data Referencing

Dynamic Data Referencing (also known as Dynamic Selection Set Functionality) allows an OpenDDA Application to modify the External Data References within the memory resident CIO to refer to different External Data values.

Dynamic Linked Library (DLL)

Dynamic linked library is a library (collection of software) that is linked to applications programs when they are loaded or run, rather than as the final phase of compilation.

Dynamic Selection Set

A Dynamic Selection Set is a collection of EDB statements whose External References will be modified in one operation by an EXEC DDA MODIFY CIO statement.

Dynamic Selection Set Functionality

Dynamic Selection Set Functionality (also known as Dynamic Data Referencing) allows an OpenDDA Application to modify the External Data References within the memory resident CIO to refer to different External Data values.

E

Economic Load Allocation (ELA)

Advanced industrial boiler control

Effective External Reference

A MapItem's iomExtRef with the IOMap's iomExtRefPrefix concatenated to the front. This is done during an I/O request just before any Substitution References are resolved.

Electric Power Research Institute (EPRI)

Located in Palo Alto, CA

Enable (Point/Unit Alarms)

A command that enables reporting of point alarms to the Universal Stations, on a point/unit basis, respectively.

Engineer

A process engineer, a control system engineer, or an application engineer. One who has a key that he or she can insert in the Universal Station key switch to gain access to engineer-accessible parameters.

Engineer Key Level

The Engineer Key Level access is required to change certain parameters (some reconfigurable through the NCF).

Engineer Key Level access can be set by using:

- A physical key in the key switch
- An assigned button
- A target in a schematic requiring a password

Engineering Units (EU)

Units of measurement such as degrees, kilos, pounds, liters, gallons, hours, meters, feet, kPa, and PSIG. Engineering units for PVs are specified as part of the data-point configuration.

Enhanced High-Performance Process Manager (EHPM)

The EHPM subsystem is a FTE-based device on the EUCN. It contains a set of algorithms and tools for implementing integrated control strategies.

The function of EHPM remains the same as the HPM.

Enhanced Network Interface Module (ENIM)

ENIM is a FTE-based node that provides interconnection between the EUCN and the LCN.

The function of ENIM remains same as the NIM.

Enhanced Programmable Logic Controller Gateway (EPLCG)

A module on the Local Control Network that links the LCN with third party PLCs.

Enhanced Universal Control Network (EUCN)

Honeywell's FTE-based, dual communications network based on ISO open system communication standards that interconnect specified Honeywell process-connected devices with one or more ENIMs.

ENIM is an interface module between an EUCN and the LCN.

Enterprise Resource Planning (ERP)

It is a software system designed to support and automate the business processes of medium and large businesses. This may include manufacturing, distribution, personnel, project management, payroll, and financials.

Entity

Entities are named collections of data that can be accessed across the LCN. There are two major entity categories: process data points and reserved entities. Reserved entities are treated differently than process data points (see the **Data Entity Builder Manual**).

A named piece of the TPS database. There are two types of entities: data points and reserved entities. Data points are built and named by users. Reserved entities are built into the system by Honeywell. Reserved entity names begin with "\$." Data point names (tag names) cannot begin with "\$."

Entity List

You can create an entity list (EL) file for use with such functions as checking or forward building in TPS Builder. An EL file is a text file, named with an extension of .EL, containing names of TPS system entities in a TPS Builder project. Each entity name is presented on a separate line in the file.

Entry Point

A subroutine in a script. The term usually applies to a subroutine that is executed by the picture runtime in response to an event.

Enumeration Type

A data type consisting of two or more states that are stored in a parameter. These states are stored in the form of strings of up to eight characters, and are referred to in these publications as a mix of capital letters and lowercase letters, for example, InActive, Active, and NotConfig are the enumeration states for certain parameters.

See Self-defining Enumeration.

For TPS Builder

The Enumeration Types function in TPS Builder refers to the creation and use of custom enumerations. An enumeration type is a data type consisting of two or more states that are stored in a parameter.

For example, if you define a custom parameter related to estimated engineering effort, you might want to define a level of difficulty as Easy, Moderate, and Complex. This could be accomplished by defining corresponding custom enumerations.

Note that the TPS Builder enumeration function does not refer to self-defined enumerations, which are configured in the STATE1 and STATE2 parameters of applicable data points.

Equipment List

A software package used in batch processes.

Equipment View

A project view that shows strategies from the perspective of equipment hierarchy: area, equipment unit, module, and points.

Error

An abnormal or undesired result of an operation or fault. A misoperation from which the hardware or software can recover. For example, memory errors that are corrected by the memory's EDAC feature, and errors in data transfers that are detected and recovered by repeating the transfer.

Establish

Entity IDs are established on the LCN at load time. This means the entity can be referenced by its external name, provided the owning node is running on the LCN. The owning node performs the external to internal name conversion to allow access to data within the entity. An entity can be established but remain not loaded because of errors or failures that prevent access to parameter data within the entity.

Eurotherm Gauging Systems (EGS)

Competitor of Lippke

Eval Subsystem

The Eval subsystem parses the data from the Fetch subsystem and converts it into a format that an application can use.

Event

An action that results in notification to some other function, for example, alarms.

Event Handler

A subroutine in a script that is called by the picture runtime in response to an event. An event handler has a distinguished name that identifies the event to be handled. An event handler is an entry point in a script.

Exception Build (EB)

For Fieldbus:

A process of creating compact files used with the UCN/LCN Data Entity Builder. The files have a .eb extension and typically contain textual descriptions of process points.

For TPS:

Use of a pre-built entity as a template and building a number of new entities, based on a text file of source data. Exception-built entities are always stored in an IDF. The template entity must reside in an IDF, either the IDF being built if the &M control line is used or another IDF if the &X control line is used, in the exception-build source code.

EXE

An executable binary file

EXEC DDA Statement

EXEC DDA Statements are syntax that you add to your OpenDDA application source code to perform OpenDDA functions at run time.

Execution Environment

For Application Director:

The software agent that manages and controls the execution of Application Director applications over a grouping of nodes in a TPS domain.

Execution_Period

For Fieldbus:

The time between initiation of block execution – for periodic execution. This time is in 1/32 milliseconds. This parameter resides in the block characteristic record.

Execution_Time

For Fieldbus:

The total time required for the block algorithm to execute. This time is in 1/32 of milliseconds. This parameter resides in the block characteristic record.

Experion PKS

A comprehensive term representing Honeywell's latest open control system that combines distributed control system (DCS) functionality and plant wide infrastructure into a Process Knowledge System (PKS)

Experion Station– Flex

A client Station that connects through an Experion PKS Server.

Experion Station – TPS

The operator station used in Experion PKS for TPS. It has the hardware (LCNP) and software to directly connect to TPN. It shares many components with the Experion Station – Console.

Experion Server – TPS

The process server used in Experion PKS for TPS. It has hardware (LCNP) and software to directly connect to a TPN. It shares many components with the Experion Server.

Experion PKS for TPS

An upgrade to the TPS system, consisting of the Experion Server and Experion Station components interfaced to TPN using TPS technology. It provides a migration path for current TPN and TPS customers Experion PKS technology. The term encompasses both the Experion Station – TPS and the Experion Server TPS.

Exploration and Production (E&P)

Upstream portion of oil business.

Exposure Rules

Parameters appearing in TPS Builder parameter configuration forms depend on exposure rules.

For example, if the type of node (NODETYP) is not entered on the form, NODENUM will be the only exposed parameter.

Another example: the number of message text items (NMSGTSXT) will be configurable only if the node type is NIM.

Expression

An arithmetic expression that can contain references to **variables**.

Extended Interface Unit (EIU)

Data Hiway to IBM Series 1

External Data

A general term that refers to data values read or written from a DCS.

External Data Block (EDB)

An External Data Block that establishes a relationship between your HLL variables and the external data values.

External Data Reference

The identification of a data value on a DCS. It consists of:

a name, which identifies the External Reference - for example, point1.paramter6(2)

a status, which indicates whether or not the External Reference represents a valid external data item

External Reference

Shorthand notation for External Data Reference

External Reference Component

External Reference Names are composed of fields called components. The name has the form POINT.PARAMETER(INDEX). The components are:

- the POINT, the part before the period
- the PARAMETER, the part after the period and before the optional index
- the INDEX, the optional part after the PARAMETER and enclosed in parenthesis

External Reference Name

The externally known ASCII identification of a data value on a DCS. For example, point1.parameter6(2).

E

External Reference Name

F

Faceplate

See GUS Faceplate

Failure

The inability of a function to perform its expected task because of a solid error or an error that recurs too frequently.

Fast Slot

In the PM-family of process controllers, fast slots are control slots are processed most frequently.

Fault

A condition, either temporary or permanent, in software, hardware, or firmware that makes a device unable to perform one or more of its specified functions.

Fault Containment

Minimizing the effects of a fault to prevent it from causing an error that leads to failure, or to prevent fault propagation to other modules (software or hardware).

Fault Tolerance

The ability of a system to maintain secure operation during the occurrence of faults.

Fault Tolerant Community

The logical system that is formed by a group of FTE nodes. Also known as community.

Fault Tolerant Ethernet

Honeywell's implementation of an NDIS intermediate driver providing automatic redirection of network traffic over multiple Ethernet ports.

FbDDService DLL

The FbDDService DLL provides a template-building interface between the DD Viewer and the Control Library.

The pathname to the Capabilities File and PD File are set in the FbDDService interface.

FbDDService input is the PD file that was generated by the DD Viewer and the Capabilities File that was created by a Device Manufacturer.

Fetch

To acquire (read) a value from a parameter.

Fetch Subsystem

The Fetch subsystem reads information out of the DD binary and converts it into a format the Eval subsystem is able to use.

FF Standard Dictionary

A file, **STANDARD.DCT**, released by the Fieldbus Foundation that contains standard strings and enumeration. A Microsoft Access Form utility compresses this text file and output a file named **IMPORT.DCT**. This file is used by the DD Viewer application.

The Standard Dictionary must be in the "Release" Directory in a compressed form.

FF Synthesizer

If only an Incremental DD exists, you must use the FF Synthesizer tool to combine a device incremental DD with FF Standard DD's to make a complete DD Binary.

The FF Synthesizer must have a **release** directory that is apart from the release directory that holds the standard and the manufacturers released DD Files. The FF Synthesizer creates its release directory during the installation process.

The FF Synthesizer release directory is similar to the *deliverable* release directory but has the following differences:

The Synthesizer adds a SYN, BIN, and a Test directory to the directory hierarchy

The Synthesizer renames the RELEASE subdirectory to GROUP during execution. Files in this directory may not be modified. Only the files in the TEST directory may be modified.

FF Tokenizer

A Foundation Fieldbus tool that converts an ASCII text file written in conformance with Device Description Language Specification into a DD binary file.

FFO file

The .FFO file extension is given to the full or complete DD binary file.

Fiber Coupler Adaptor/Repeater or Fiber to Copper Adaptor/Repeater

Provides a way to electrically isolate and extend a serial link in the Field I/O system. The Adaptor feature converts light energy to electrical energy. The Repeater feature allows extending the serial link optically.

Fiber Optic Cable

Two or more glass fibers wrapped and jacketed for inside, outside aerial, or outside direct burial use.

F

Fiber Optic Clock Receiver (FOC/RCVR)

Fiber Optic Clock Receiver (FOC/RCVR)

I/O-size board that terminates a fiber optic segment for LCN clock transmission for a system with non-KxLCN processors in the remote segments.

Fiber Optic Clock Transmitter (FOC/XMTR)

I/O-size board that transmits through a fiber optic segment for LCN clock transmission for a system with non-KxLCN processors in the remote segments.

Fiber Optic Link Segment

A fiber optic cable with at least two fibers that interconnect LCN Extension Set boards. Each of the two fibers transmits in the opposite direction.

Field Device

For Fieldbus:

Foundation-compatible device that contains and executes function blocks.

Fieldbus (FB or FBUS)

The emerging standard for digital communications among field instruments and control systems. Also known as IEC/ISA Fieldbus.

Fieldbus Foundation (FF)

An independent, not-for-profit organization that supports a single, open, interoperable fieldbus technology based on the work of the IEC and ISA.

Fieldbus Link View

For Fieldbus/TPS Builder:

The layout of a fieldbus link used as a basis for organizing strategies (that is, link, devices, and function blocks/transducer blocks). TPS Builder allows a project to be looked at from these perspectives:

- Equipment View
- Hardware View
- Control View
- Fieldbus Link View

Fieldbus Message Specification (FMS)

Specifies the message structures that communicate between the application layer of the device and the Fieldbus Access Sublayer, which communicates with the Data Link Layer.

A number of FMS services are defined. Field devices may support all or a subset of the FMS services. One of the most frequently used services is the FMS Read service that allows the reading of a parameter from a remote device.

File Name

For TPN:

The eight character name of a file plus its suffix.

Firmware

Software whose ultimate source is nonvolatile memory

First Match Group

A window group in which the search for a matching subgroup and/or window specification is always top-down, depth-first, according to the physical ordering of the information in the workspace configuration file.

Five/Ten Slot

A module that has either five or ten card slots that accommodate the boards required to function as a node.

Flag

Flag is type of Data Point found in different control devices such as AMs, xPMs, and MCs.

Format

Typical use of this term is for the operation of formatting disks, cartridges, or HMs. Formatting consists of writing track headers, establishing the volume name for a cartridge, and building the bad-sector table for the HM.

Formatting destroys any previous data on the medium.

Forms Management System (FMS)

Part of VAX software, EH800

Forward Build

Forward building is the process of creating a text file, named with an extension of .EB, of TPS system entity (node and point) parameter data in a TPS Builder project. The EB file follows the syntactic and format specifications required by the TPS system platform.

Free On Board (FOB)

Freight and insurance responsibility

FTE Community

An FTE Community is a collection of FTE and non-FTE nodes that are able to communicate with one another.

FTE Cluster

An FTE Cluster is a group of nodes with high intercommunication in which all the nodes are connected to one switch pair. Each cluster requires a pair of redundant switches - Switch A and Switch B. Each of the cluster switches connects to its corresponding backbone switch (Switch A or Switch B).

FTE Tree

An FTE Tree is formed by connecting each FTE Node in the network to both switches in the redundant pair. All of the A ports on the dual adapter are connected to one switch (A switch) - all of the B ports on the dual adapter are connected to another switch (B switch). The A ports and A switches form **Tree A**; the B ports and B switches form **Tree B**. The separate identity of the two FTE Trees is maintained by using the pre-configured color-coded cables.

FTE Node

FTE Nodes are those with the necessary redundant media components and Honeywell FTE software. FTE Nodes connect to the LAN using dual port NIC adapters; each adapter port has a unique IP address. FTE Nodes are resilient to single Ethernet failures such as switch or cable faults, and will be able to communicate as long as at least one path exists between them.

Function Block (FB)

Executable software objects that perform a specific task, such as measurement or control, with inputs and outputs that connect to other entities in a standard way.

Function Block Application Process (FBAP)

Consists of one Resource Block, zero to several Function Blocks and zero to several Transducer Blocks. Another name for the FBAP is an application VFD (Virtual Field Device).

Function Block Control Module (FBCM)

A subset of the TPS Network CM (Control Module) point type. The FBCM represents function blocks and transducer blocks within a Fieldbus device on a link.

Function Block Schedule

Within a device is a portion of the link-wide schedule. The execution time of a block is represented as an offset from the beginning of a device's macrocycle.

Function Block Start Entry Objects

This object represents an entry into the function block schedule of a device. The object consists of a start time offset from the start of the macrocycle, the function block object index, and a VFD reference.

Function Set

A named collection of software procedures and tasks that provides a particular function or functions.

G

Gain (K)

The proportionality constant of a controller. The gain determines the change in the valve opening corresponding to a given error.

Gateway

A specific type of module on the TPS Network that converts the communication protocol of other networks to and from that used by the TPN. The terms **module** and **gateway** usually represent a TPN system component that is connected to the TPN. The term **box** always represents a TPN process-connected device that is connected to a Data Hiway.

General Output Connections

Optional point interconnections (Application Module and Process Manager/Advanced Process Manager points) used to store (or push) the output values of any local point to named destinations that are usually external to the point.

General Output File

A file OpenDDA creates to contain the general output of an executed application. This includes any normal printing statements of an application, along with any other application output.

Generic

A term applied to applications and application packages that indicates that the application being discussed applies to any of a particular kind of equipment, equipment grouping, or product.

Generic Data eXchange (GDX)

GDX is a Honeywell set of software providing the exchange of data between the Host platform (HP-UX) and the DCS (TDC3000x).

Generic OPC

COM DLL that provides the generic implementation of the OPC data access for those HCI Servers that choose to utilize OPC

Generic Parameter

For Fieldbus:

Parameters which are generally characterized. For example, input parameters could be one generic class, and output parameters another. Or, generic parameters may be more specific, such as based on data type such as input parameters of type floating point.

Gigabyte (GB or Gb)

Equals 1,073,741,824 bytes

Global Application Module on NT (GANT)

An AM and Windows NT combination node which is planned to be sold only as part of a solution.

Global DDB

One of the two display databases on a Universal Station. The data in the Global DDB persists between picture invocations. The Global DDB can be used to pass data from one picture to the next.

Global Description File (GDF)

The GDFs are the master description files used to describe the parameters and options available at entity-build time. The GDFs contain all the build-parameter descriptions and the forms (displays) presented to you in the parameter entry displays. There is a GDF for each type of entity.

Global Output Focus

Only one window in the entire workspace can have output focus status at any one time and it overrides any group-level **output focus** windows in multiple-group workspaces.

Global Resource Center (GRC)

Services business

Global User Station (GUS)

Global User Station is an intuitive and productive human interface to the Honeywell TPS system. This station makes plant-wide information easily accessible for needed business and control information. GUS is a PC-based system using Microsoft Windows NT and an LCN Native Window interface.

Globally Unique Identifier

A 16-byte value guaranteed to be unique for all programs, on all machines, for all time

Graphic

A symbolic shape or collection of shapes which forms a recognizable image. Also see Schematic.

Graphical Connection

The method used for representing a data flow connection between two blocks that are on the same control drawing. A line connects the two blocks.

Graphical Workspace Editor (GWE)

For SafeView:

A SafeView application that provides a graphical human interface for creating and editing workspace configuration files. The Graphical Workspace Editor (GWE) provides displays of the structure of a workspace configuration, including its component window groups and window specifications. Each workspace component can be viewed and edited using the tabbed property page dialogs. There is no API programmatic interface provided by the GWE.

G

Greenwich Mean Time (GMT)

Greenwich Mean Time (GMT)

The mean solar time of the meridian of Greenwich, England used as the prime basis of standard time throughout the world.

Same as Universal Coordinated Time (UCT).

Grid

The border on a printed control drawing, with labeled horizontal and vertical divisions. Used for identifying and locating off-page connectors.

Group Display

For GUS:

An NT-hosted display that is functionally equivalent to the Universal Station R520 Group Display.

For TPS:

A display of up to 8 points with bars and values used for control.

See Operating Group.

For SafeView:

See Window Group.

Guest

For Remote GUS:

During Remote Control, the PC connecting to and controlling another PC.

GUS Application

An application built to run on GUS.

GUS Basic

The scripting language used in GUS picture scripting. It is similar to Microsoft Visual Basic.

GUS Client (GC)

A computer capable of executing Windows NT 4.0 and connected to a single GUS Server through a communication link. It is capable of hosting the Client components required to achieve Remote Displays.

GUS Display

A display created using the Honeywell Global User Station (GUS) Display Builder software. Typically used for operational viewing and control of distributed control system processes.

GUS Display Builder

An application for creating process control displays that can access data through both HOPC and HCI/OPC.

GUS Faceplate

An OCX representing one “slot” of the GUS Group Display.

GUS Keylock

The security level associated with data access requests between a GUS node and the LCN. This can be manually changed on a GUS Integrated Keyboard (IKB) through a physical key setting, or through a GUS script, or through the Native Window. The three levels for the GUS Keylock are: Operator, Engineer, and Supervisor.

GUS Mode

The mode (for example, *GUSRemote*, *GUSEmbedded*) in which a GUS display may be found to be executing. The GUS scripting function GetGUSMode returns the current GUS mode.

GUS Node

A W2K capable computer with an LCNP card executing the GUS personality.

GUS Picture Builder

Also known as GUS Display Builder. An application for creating process control displays that access both TPN and HCI data.

GUS Remote Cluster (GRC)

The set of computers consisting of a single GUS Server and up to four associated GUS Clients.

GUS Server (GS)

An NT-capable computer connected to the TPN through an LCNP. The LCNP executes the Universal Personality.

A GUS Server can be connected to 1 – 4 GUS Clients.

H

Hardware View

The Hardware View is a project view that shows a hierarchy of network, node, and points.

HCI Managed Component

A software entity that can be named and established in a manner compliant with the TPS system. Application Director is a managed component of the TPS system and is visible in the TPS System Status display.

HCI/OPC Name Form

The name form used by display authors to access HCI/OPC data; for example, HCI.<serverAlias>.<dataItemName>.

HCI/OPC Surrogate Server

An HCI/OPC server nominated as an alternative data source for GUS display data referenced through a given HCI/OPC server. If multiple HCI/OPC servers are referenced in GUS displays, multiple surrogate servers may be provided.

HCI Server

A data server within the TPS system, which is compliant with the Honeywell Communications Interface (HCI). The TPN Server product is an HCI server.

Heartbeat Provider

Heartbeat Provider represents a WMI Provider that sends event notifications to WMI clients when a node changes state.

Honeywell's Fault Tolerant Ethernet WMI provider. Provides enumeration and instance creation and deletion for nodes taking part in the FTE community configured on the local node.

Heartbeat Server

Refers to the combination of the Heartbeat Provider and Heartbeat Client connection. Used to notify the local node of the online status of remote nodes.

Hertz (Hz)

The unit of frequency of one cycle per second.

High Level Data Link Controller (HDLC)

Computer protocol

High Level Language (HLL)

Computer languages (such as Fortran or C/C++) used to write applications.

High Temperature Short Time recorder (HTST)

Dairy application

High-Performance Process Manager (HPM)

A process-connected device on the UCN that provides regulatory control and sequence operations. The HPM is faster than the APM yet includes all of the data acquisition and control algorithms in the APM, plus additional algorithms and features.

History Module (HM)

A node that provides mass storage. History Modules (HMs) are principally used to store historical data including continuous history, and various journals and logs. They can also be used to store software, system files, and user files. HMs are also used to load other LCN nodes.

Historians

Subsystem responsible for collection and retention of process variables and events

History Module Initialization (HMI)

The History Module Initialization personality used for NCF changes, running EC files, and backing up history files.

History Module Operating (HMO) personality

The History Module Online personality

Hiway

The old TDC 2000 Data Hiway, which has several types of boxes, connected to the process such as: CBs, ECs, MCs, HLPIUs, and DHPs.

Hiway Gateway (HG)

A node that interconnects the Data Hiway to the LCN.

HM Load

The operation of loading the system software to the HM(s).

HM Load also refers to how heavily loaded the HM is with history groups or other resources effecting CPUFREE (checkpointing, backups, file access).

H

Honeywell Communications Interface(HCI)

Honeywell Communications Interface(HCI)

The set of COM interfaces and client libraries that provide the API for communications portion of the System Infrastructure.

HOPC Name Form

The name form used by GUS display authors to access LCN data through HOPC; that is, LCN.<point>.<parameter>.

Host

For Remote GUS:

During Remote Control, the PC being controlled by another PC.

Host Platform

The Host Platform is a hardware and software computer system that serves as an application development environment for OpenDDA. Host Platforms supported by OpenDDA include:

Honeywell's Application Module x (A x M), which includes Hewlett-Packard's PA-RISC coprocessor running HP-UX operating system version 9.05 or later

A Hewlett-Packard workstation based on Hewlett-Packard's PA-RISC coprocessor running HP-UX operating system version 9.05 or later

Hotkeys

For SafeView

A Safeview hotkey provides direct access to specific functions regardless of which application has user input focus. Hotkeys are provided to:

- show/hide the standard workspace control panel,
- invoke the Display History dialog,
- reposition all application displays to their original positions, and
- invoke the prior display (the most recent entry in the Display History dialog).

HREF

An HTML attribute that sets or retrieves the destination URL or anchor point.

HRESULT

A COM and OPC common structure specification for status returns.

H
HRESULT

ICON Console

A Honeywell product providing GUS node functionality on an advanced user interface platform providing up to four physical screens.

IKB Service

Executable module implemented as Windows NT service. Installed at the PC that has the GUS Integrated Keyboard (IKB).

Image Point

An operating point in the CLM that images one or more registers and coils in a Modbus device.

IMPORT.DCT File

The FF Standard Dictionary is compressed by a utility module and named IMPORT.DCT. This file is used by the DD Viewer application.

In-Proc

“Within the same process”; usually referred to with reference to a DLL

Inconsistency Error

An inconsistency error occurs when OpenDDA detects a mismatch in data size or array size between the application variable and the value being moved into or out of the application. An example is when the definition of an external data item in the CIO file does not match the corresponding HLL variable definition in data size and array size.

Indisposable

See Disposable.

INF

A text file, bundled with a CAB file, which specifies the instructions for Internet Explorer to use to download and set up an ActiveX control.

Infinite

A value that can be configured for the maxwindows window specification property. If maxwindow = infinite in a window specification, this specification can be simultaneously used to manage an unlimited number of actual displays.

Inherit Background

A WebGUS Property indicating whether the WebGUS runtime display background should reflect the containing document's ambient background color.

Inhibit

For Point/Unit alarms

A command that suppresses detection of point alarms to the Universal Stations, on a point/unit basis, respectively.

Initializable Inputs

Input parameters of control algorithms through which initialization can be passed upstream if the control algorithm are configured for external initialization. The setpoint of a PID is an initializable input.

Initialization

Adjustment of values in a cascade strategy when the cascade is open, such that when the cascade is again closed, a process bump does not occur.

Initialize

For an HM, initialize is establishing the volumes that exist on the HM.

The volume directories are cleared; so all data previously available can no longer be accessed and must be restored. This term is sometimes used in place of format when referring to cartridges. The utilities support initializing the removable media (clearing the directory and renaming the volume) without formatting (writing track headers).

Input/Output (I/O)

You can specify/control when I/O occurs to the data platform. HLL variables are populated/depopulated by generated code within the body of an application.

Input/Output Module (IOM)

Input/Output Module contains the data points in an I/O Processor (IOP) that perform input and output processing of all field I/O. IOP data points perform these functions independently of the control functions.

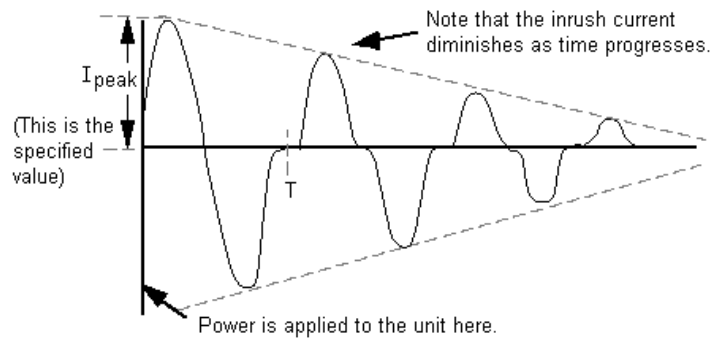
I/O points are classified as Analog Input, Analog Output, Digital Input and Digital Output, but have further module types within these main classifications to describe the type of IOP exactly. An IOP contains 8, 16 or 32 data points.

Input/Output Processor (IOP)

The TPS controller nodes (PM, APM, HPM, and LM) have Input/Output Processor (IOP) boards. Both the primary and backup IOPs receive all data from the UCN or from the process simultaneously, and should the primary IOP fail, the backup IOP takes over automatically, becoming the new primary IOP.

Inrush Current

This is the peak current that occurs when power is first applied to a unit. It is measured in peak amperes as shown below.



Insertion Point

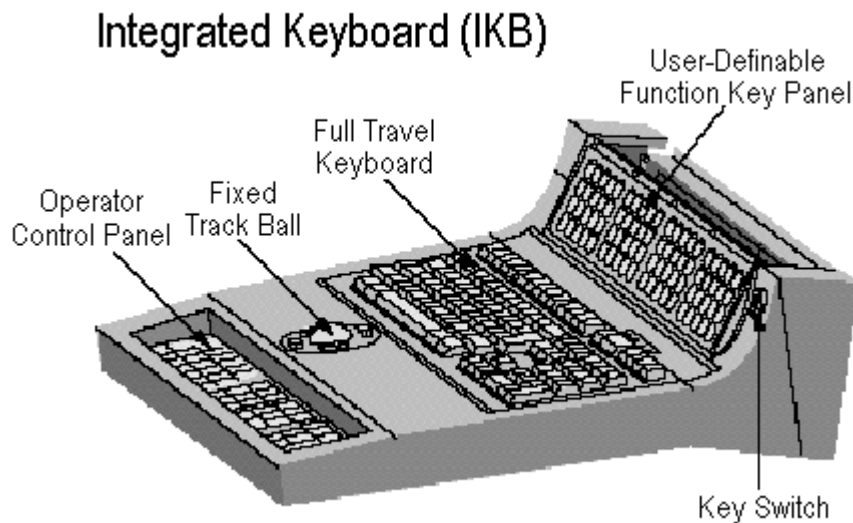
A place in the processing order of a data point in an Application Module at which execution of a Control Language block is initiated. When execution of the block is complete, processing of the data point resumes immediately after the place of insertion.

Install Application

OpenDDA command line function which prepares an OpenDDA Application for online execution.

Integrated Keyboard (IKB)

A specially designed keyboard that can be hooked to a GUS Server and/or GUS Clients to support GUS operations.



Integrated Services Digital Network (ISDN)

An evolving set of communications standards for a digital network carrying both voice and data.

Intermediate Data File (IDF)

The IDF is a file used by the Data Entity Builder to store the build information for an entity. It is not the complete entity description, but contains the data necessary to support loading to the destination node.

Internal Id

DCS specific internal representation for an External Reference Name. Typically, accessing data using the Internal Id is much faster than using the equivalent External Reference Name.

Interoperable Systems Project (ISP)

Fisher, Rosemont, Siemens, Yokogawa fieldbus

Invocation

Action to cause a display to bring up on a screen

Invoke

This is the action of calling up a specific function or display on the US.

Typically used for describing the action of calling up a schematic display.

I/O Control Module (IOCM)

This module coordinates communications between the Logic Controller processor and I/O system and formats the data flowing between the I/O system and the processor. It also monitors individual I/O module fault diagnostics. The IOCM works in conjunction with the Parallel Link Driver Module and/or Serial Link Module to control data flow to I/O.

IOLBIM

Bus Interface Module for the I/O Link. It is a multi-board module consisting of at least the Rail Bus Processor or “RBP” board, an Application Processor Board or “APB” and an IOL-specific “LAN” board.

IOMap

A collection of "MapItems" (either in an IOMap file or defined in a program), used to configure an IOMap Server.

IOMaps allow you to build applications using "generic" terms for controller-related input/output, such as "FLOW" instead of a specific flow point.parameter accessed through a specific server. Use of generic terms streamlines your application development. To run an application, an IOMap is used to correlate the generic terms to a specific inputs/outputs of interest.

IOMap Component

A term used to refer to IOMap Server, IOMap, and MapItem.

IOMap (Map) File

An IOMap saved to a file, by use of either the iomSave property, or TPS Builder. An extension of ".iom" is used for these files.

Map files may reside anywhere in the file structure of a TPS system, as designated by the application developer. If the developer stores these files in system-specified locations, the TPS system infrastructure can provide certain benefits (such as security and replication).

IOMap Property

A term for atomically readable and writeable elements of IOMap components.

IOMap Properties are specific characteristics (readable and writeable elements) of a MapItem, IOMap, or IOMap Server. Examples are Name, Cross Reference, Test Value, Test Mode, and External Reference (maps correlate "generic" names to specific external references).

IOMap Server

An in-process server, offering services through Microsoft COM interfaces. When used in concert with Honeywell Communication Interface (HCI), this server offers specific interfaces conforming to the OPC specifications for data access. IOMap Servers contain IOMaps.

IOMap Text File

A UNICODE text file created by the IOMap Convert Tool, or TPS Builder's Export function. Text files present the contents of an IOMap in a user-readable format. The Convert Tool and TPS Builder present the information in slightly different formats.

I/O Processors

Components in the I/O subsystem of the Process Manager and Advanced Process Manager that are connected to the process through Field Termination Assemblies. Included are :

- High Level Analog Input
- Low Level Analog Input
- Low Level Multiplexor Analog Input
- Smart Transmitter Interface
- Serial Device Interface
- Analog Output
- Pulse Input
- Digital Input
- Digital Input Sequence of Events
- Digital Output processors

IOP A

IOP A is the physical primary IOP. It is the IOP that is physically connected by cable to the “A” cable connector on a redundant FTA on xPM nodes in the UCN.

IOP B

IOP B is the physical primary IOP. It is the IOP that is physically connected by cable to the “B” cable connector on a redundant FTA on xPM nodes in the UCN.

IOP Bias

For the 8-channel Analog Output IOP only, operation with the IOP A as the primary IOP and IOP B as the secondary IOP is with the basis, the preferred condition. Operation with the IOP B as the primary IOP and IOP A as the secondary IOP is against the basis, the nonpreferred condition.

IOP Swap

An IOP swap reverses the status of the physical IOPs, A and B. The present primary IOP becomes the secondary IOP, and the present secondary IOP becomes the primary IOP.

J

Journal

A list (optionally stored on the History Module) that records all significant events of a particular type (for example, process alarms, operator process changes, and system maintenance messages).

JS File

A source text file that holds configuration information for Multifunction Controller or Advanced Multifunction Controller (MC/AMC) logic blocks. The file content follows a Honeywell-defined format.

Jumper

A movable device for providing an electrical connection or short between to points on a circuit board for the purpose configuring the equipment for a specific function or operation.

J
Jumper

K

K2LCN

LCN module printed circuit board that contains a microprocessor, memory, and LCN interface.

Keyword

A reserved word in the workspace definition language. Keywords describe specific kinds of information involved in configuration workspaces (.WDL files).

Kilobyte (KB or Kb)

Equals 1024 bytes

K
Kilobyte (KB or Kb)

L

Label Attribute

The Label attribute of a parameter approximates the name but is not the name. The Label attribute may be composed of several words separated by blanks.

The Label attribute is a field to be displayed near the value of a parameter on an operator console.

LCN Extender board (LCNE or LCNE2)

Single board (I/O size) that interconnects one coaxial cable in a coaxial segment to a single full-duplex fiber optic cable fiber.

LCN Extension Set

A set of boards that extends the LCN through the use of fiber optic cable.

LCN Fiber Link (LCNFL)

LCN Fiber Link board that terminates two Fiber Optic cables (Cable A and Cable B), and is used when no clock is needed.

LCN Node Address

A two-digit number that uniquely identifies a node such as a US on a TPS Local Control Network.

LCN Surrogate Server

An HCI/OPC server nominated to serve as a data source for data referenced in a GUS display using the HOPC ("LCN.") Name Form. For multiple-LCN systems, multiple LCN surrogate servers may be configured.

Level 3 System

A term used to describe the set of products that enable efficient production, operation, and maintenance of supervisory and advanced control applications.

Levels of Control

Three levels of control are sometimes referenced when considering complex total control strategies. In today's world the boundaries between these levels are somewhat arbitrary.

Level 1 is the process-connected level, where control functions are normally process-hardware oriented. Single-loop regulatory, digital-loop, and sequential control are all implemented at this level in process-connected devices. Control strategies to meet most process requirements can be implemented at this level with a very high degree of process reliability.

With TPN, all Level 1 control functions except sequential control are implemented by configuration. Sequential control is implemented through a Honeywell-designed process-oriented programming language.

Level 2 control is process-unit oriented. At this level, strategies for unit control and unit alarming can be implemented. With TDC 3000X, most unit strategies will be implemented by configuration, but application-specific problems can be solved using control language.

Level 3 control is process-unit or process-area oriented. At this level, unit- or plant-wide strategies and optimization routines that require linear programs or other computing capabilities can be implemented. This level is implemented through a high-level programming language, such as Fortran or Pascal.

Library

Libraries are convenient template managers that enable you to view, access, and organize templates. Honeywell provides system libraries, and you can also create your own. Separate subsystems, that is sets of libraries, can be purchased depending on your configuration needs.

Light Emitting Diode (LED)

A semiconductor device emitting incoherent optical radiation when biased in the forward direction.

Most boards have LEDs to indicate if they passed self-test or have an error. The processor boards also show LCN address when normal and an error code when not normal.

Link

For Fieldbus:

A single circuit of Fieldbus devices, which are assigned node addresses on the link. One single active link master device on the link distributes data link time and link schedule time to the nodes. Link and Segment are used interchangeably in Fieldbus.

For CL:

The operation of binding final components into the system. Typically, this is used to describe the CL-link operation where a CL-object program is linked to the entity(s) that uses it. Any indirect entity reference, which was not bound at compile time, must be bound at link time. This means the entity must be built and loaded, and the owning node must be running on the LCN, at compile time.

Link Active Scheduler (LAS)

For Fieldbus:

An entity in a device on the link that serves as the link master. The LAS removes non-responding devices from the link and recognizes new devices that are coming on to the link. The LAS administrates the active link schedule by transmitting a CD token to the data link buffer located at a publisher's VCR endpoint.

Link Master

For Fieldbus:

A class of device that contains a Link Active Schedule. Each H1 Fieldbus network has a device that is the Link Master and optionally one or more devices that are capable of assuming the role of backup to the Link Master. In the TPS Fieldbus system, the Fieldbus IOP is configured as the Primary Link Master. A device such as the Honeywell ST3000 can be configured to assume the role of the Backup Link Master.

Link Object

For Fieldbus:

Contains information needed to define connections between function blocks. Link objects also define connections between interface devices and field devices.

A link object consists of a local index, a VCR (Virtual Communications Relationship) number, a remote index, a service operation, and a stale count. Link objects reference publisher-subscriber or client-server VCRs.

Link Schedule

For Fieldbus:

Defines the time at which published parameters of function blocks are scheduled. The combination of the link objects for a link and the function block schedules are used to construct link schedules. A link schedule(s) resides in a domain within the NMIB of a LAS device.

Link Units

Address a constraint that arises in the HPM. Whereas node configuration in the PM and APM are constrained by Processing Units, the significantly increased processing capacity of the HPM means that the constraint on HPM performance can often shift to I/O link access.

Link access can be thought of as I/O bandwidth, and link units as the measure of the capacity of I/O link access.

TPS Builder Link Units allow you to evaluate various configurations. You can then revise your HPM node configuration based on the HPM link units calculations.

Load

Used for several operations in the system:

- Loading a specific node's memory with a personality image from a removable medium or HM
- Loading an entity, using the DEB, to its destination node (or file in the case of the area database)
- Loading an area database from its file to memory when a US is loaded and started up
- Loading a checkpointed database from the checkpoint file to the memory in an HG or AM

Local Area Transport (LAT)

Ethernet communication

Local Control Network (LCN)

The TDC3000 communications network that connects multiple nodes. A US/GUS is one such node type.

The Local Control Network interconnects the LCN-based nodes. The LCN employs dual, redundant cables, cable A and cable B.

The link through which all TDC3000-compliant nodes communicate. The term is often used to generically name the link with the nodes attached—the system.

Local Control Network Extender (LCNE)

A Local Control Network device that serves as the interface between coaxial cable and fiber optic cable. This allows remotely placing LCN module(s), up to 2,000 meters from the Local Control Network.

Local Control Network Processor board (LCNP or LCNP4)

Board that controls the communication with the TotalPlant Solution (TPS) Local Control Network (LCN). LCNP(4) boards are found in GUS, APP, and PHD nodes.

Local DDB

One of the two display databases on a Universal Station. The data in the Local DDB does not persist between picture invocations. In other words, data, the local DDB, is initialized as empty when a picture is invoked.

Local LCN

Term used to differentiate a host LCN from other (remote) systems during PIN planning and configuration. The PIN configuration information is entered separately at each individual LCN, thus each LCN is local for one configuration session and a remote for all others.

A Network Gateway (NG) or TPN Bridge is used to communicate between the local and remote LCNs.

L

Local-Only Component

Local-Only Component

A component that is always accessed by local clients and typically has the same alias name on multiple nodes. Its alias name cannot be duplicated on the same node. A local component's alias name also cannot be a Domain Component's alias name.

Local System Number

A two-digit number (01-64) that uniquely identifies an LCN system on a PIN for use with Network Gateways (NG) and Global User Stations (GUS).

Localization

Modification of system to use local language

Lock Button

An indicator button in the title area of round robin grouped windows that allows the operator to exempt a window from the round robin automatic display replacement algorithm.

The operator cannot lock all of the windows in a round robin group; if only one window remains unlocked, its lock button is disabled.

Logic

Logic Data Point (Slot) found in xPM nodes.

Logic Manager (LM)

A process-connected device on a UCN that provides high-speed logic functions, including fast ladder logic program execution, extensive digital, boolean, or interlock logic, and ease of ladder logic programming by the user.

Logic Simulation

This TPS Builder function allows you to simulate the logic flow of a TPS Network Logic or Device Control point, represented in TPS Builder as a template or strategy module.

In simulation, you can set logical inputs, flags, and numerics. You can run a simulation on a fully configured point, or begin with simple logic configured, run the simulation, add more to the logic, run the simulation again, and so on.

Logical Console

Any set of up to 10 Universal Stations and/or Universal Work Stations defined as a "console" during system configuration. This can consist of stations that are physically adjacent or physically separated and/or loaded with different sets of software.

Logical Device Identifier (LDID)

This is a name by which cartridge and printer devices can be accessed on a console through the utilities. The LDID is of the form \$Fn for cartridges, and \$Pn for the printer (n is 1-20).

Logical Hiway

That part of a Data Hiway that consists of all boxes interfaced to the TPN through a given Hiway Gateway or Hiway Gateway redundant pair. Another Hiway Gateway (or redundant pair) interfacing different boxes on the same physical hiway is considered to be serving a different Logical Hiway.

Logical Link

The logical link expresses the capability of two systems to exchange information through the communication network. A logical link may include several physical links.

Logical Node

A module on the Local Control Network that generally is treated as a single device, although it may physically consist of more than one device—a primary, plus backup.

Logs

TPS

Several log types with different formats such as Free Format Logs, are available through the Area Data Base (ADB) to retrieve current or history information for points.

TPS Builder

TPS Builder maintains a log of important processes performed within a project. These processes, such as Address Assignment, add a message to the log showing the date and time the process was executed, the user who performed the process and if the process completed satisfactorily or if errors were encountered. The log is maintained for each TPS Builder project. TPS Builder provides functions to view and maintain the log.

Long Word

Data type that equals 32 bits, 4 bytes, or two words

M

Macrocycle Duration

For Fieldbus:

This attribute in the NMIB (Network Management Information Base) specifies the duration of a schedule in 1/32-millisecond increments.

Managed Node

A managed node represents a node that is WMI enabled and has the WMI providers installed.

A node containing System Management runtime software. The node is capable of supporting the System Management Display

Manual Select Group

A kind of window group in which the window of output focus is determined only by specifically selecting the window's option focus button; this window represents the output focus (where the next application display will be presented) until the operator manually selects a different window. It is important to note that a workspace can consist of multiple window groups, each of which is typically intended to contain a related set of application displays. Each window group can have its own window of output focus. Thus, a workspace with multiple workspace window groups will have multiple windows of output focus.

See Global Output Focus.

Manufacturing Facilities Integrator (MFI)

Modified CLM

Manufacturing Operations Management (MOM)

From the Gartner Group

Map Files

IOMaps can be saved into files. Complete map data, including all property definitions, are contained in this file.

MapItem

A MapItem is a pre-defined collection of properties. You define MapItems during IOMap creation.

Match Expression

An expression that is evaluated by the SafeView at runtime to determine if an associated workspace group/window specification should be used to manage an application display. A match expression can express any combination of the following kinds of values: *module* (file path), window title, and *window category*.

Maximum Value

The current values shown as maximum (Max.) in the tables were measured with a load approximately 30% higher than the load used for measurement of typical values to account for variations in components. Also, these measurements were taken with the minimum rated voltage, 102 Vac RMS, applied to the supply.

Maxwindows

A property available to window specifications only for first match window groups. This property determines whether the window specification can be simultaneously applied to an unlimited set of application windows or just one application window. This property is normally 1, but can be set to *infinite*. Such a setting is useful when, for example, a single window specification is configured to catch all miscellaneous applications that are not specifically managed elsewhere in a workspace configuration.

Megabyte (MB or Mb)

Equals 1,048,576 bytes

Megahertz (MHz)

A unit of frequency of 1,000,000 hertz

Megaword (Mw)

1 Mw = 2 MB (Megabytes)

Memory Module

The Logic Manager component that holds the ladder logic program.

Meta-data

For Fieldbus:

TPS Network parameter attribute/display format extension to the various Fieldbus Foundation (FF) defined view parameter sets of a fieldbus device. This information is produced by the TPS Builder for a TPS Network CM point which allows a mix of standard FF device data, and non-standard manufacturing data to be projected, mapped or displayed in TDC, primarily for display purposes. Simply put, Meta data is “data that explains data.”

Methyl Tertiary-Butyl Ether (MTBE)

Gasoline oxygenator

Micron

Micron is another term for micrometer, a unit of measure, one thousandth of a millimeter or one millionth of a meter.

The symbol μm is often used to denote micron or micrometer.

The abbreviation nm denotes nanometer (one billionth of a meter) and is used as a unit of measurement for the wavelength of light.

Migrate

Movement of configured items and applications from one control system to another. For example, control schemes from TDC 3000X to TPS

Mode

The characteristic of a data point that determines how the data point operates; for example, automatic mode for a Regulatory Control Point specifies that the point's setpoint is to come from the Universal Station, while the point calculates its output according to its configured algorithm. Manual mode, on the other hand, instructs the point not to execute its algorithm and to accept an output value directly from an external source, such as the Universal Station or a program in a module or process-connected device.

Mode Attribute

A parameter of output data points that determines whether an operator or a program can change a point's mode.

Modified Accelerated Cost Recovery System (MACRS)

Equipment depreciation

Module

TPS Builder

A template or strategy composed of other strategy blocks and/or modules. For example, you might create a module to represent a flow loop. A module is represented in a control drawing.

TPS

A unit containing power supply, fans, and backplane that, with a compliment of boards, performs a specific TotalPlant Solution (TPS) System function. For example, any node on the LCN such as a US, HM, or NIM.

SafeView

The file name and path of an executable file name. An application's module is one of the three means by which an application's display window can be matched and thus managed by SafeView. The other two are **window title** and **window category**.

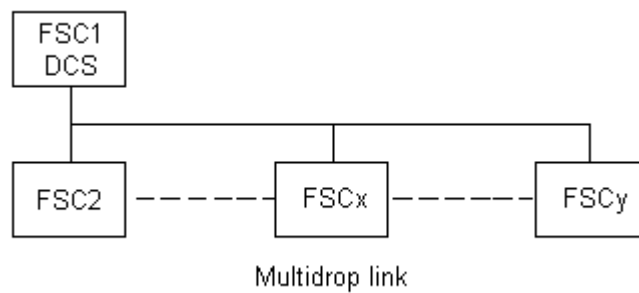
Module Generation

A TPS Builder function that allows you to populate an existing module ("empty" or containing other strategies) nearly automatically.

This function works if you your project includes a number of strategies for which connection relationships have been configured. The generation is based on the connection relationships. This is particularly useful when many TPS Network points have been or will be backbuilt as strategy blocks into TPS Builder.

Multidrop Link

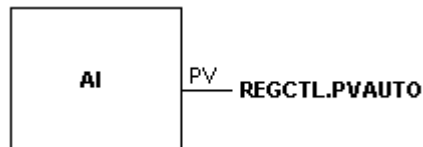
A multidrop link is a physical link, which interconnects multiple systems. Within the context of the FSC communication, a multidrop link is the connection of a single master system with multiple slave systems.



N

Named Connection

In a drawing, a named connection presents the name of a data flow connection, rather than using a line between two drawing symbols. The connection line goes from the block that is on the drawing to a label that displays the unique name and parameter for the block on the other side of the connection:



Whenever connections span multiple pages in a printed drawing, a named connection is used.

Named Data Access

GUS Named Data Access supports both HOPC-based and HCI/OPC-based data items. It is the ability in the GUS Display Builder to access server data items by name without having to specify the location of the server, and for HCI, without having to make OPC-specific calls.

National Electrical Code (NEC)

United States standard for NEC electrical installations

National Electrical Manufacturers Association (NEMA)

An association of NEMA electrical equipment manufacturers that establishes consensus standards through its membership.

There are different types of NEMA boxes to protect field termination from the environment.

National Fire Protection Association (NFPA)

Sponsor of the National Electrical Code

Native Window

The native window is a GUS display window that provides a direct user interface with the LCN. This particular application window is the only means by which LCN standard displays, such as the classic group display, can be viewed. It is an example of an application window that can be managed by SafeView.

Network Configuration File (NCF)

The NCF contains the description of the system, so that each node is aware of the system structure. The NCF is time stamped to the millisecond and all nodes have to be loaded with the current version.

- Online NCF changes should be made to avoid reloading all nodes at the same time or to avoid an LCN reconnect procedure.
- Offline NCF changes are used to change Unit IDs or position, and from some NG configuration parameters.

Network Definition Table (NDT)

The NDT contains the definition of every network component within the CLM realm, including: the CLM Interface board(s), the ports on each interface board, and each addressable component in the non-TPN network (for example., gateway, bus, device subdevice). In addition, the NDT contains the relationship between these network components.

Network Device

Network Device refers to the non-TPN device or network that is interfaced to the TPN through the CLM.

Network Gateway (NG)

The node that logically connects two LCNs together. Currently, the list of supported remote systems is limited to TPS Local Control Networks. The NG interconnects with other systems through an open system Plant Information Network (PIN) running an IEEE 802.4 protocol on either a 5 Mb/s Carrierband network, a 10 Mb/s Broadband network, or a 10 Mb/s fiber optic network.

The Network Gateway name (also referred to as the LCN name) is prefixed to the point name when used in an External Reference Name. An example is:
SS\POINT1.PARAMETER5 where SS is the LCN (or Network Gateway) name. The LCN name can also be used for local LCN access when there is no Network Gateway as long as the local LCN is configured with a name.

Network Interface Module (NIM)

The node that provides the interconnection between the Universal Control Network (UCN) and the LCN containing xPM and LM nodes and their point databases.

Network Management Information Base (NMIB)

For Fieldbus:

A collection of variables associated with Network Management. The NMIB contains configuration, performance, and fault-related information for the communication system of the field device.

Network Memory

A segment of the CLM Node's redundant memory allocated for shared access between the CLM and the interface board to facilitate CLM network communications processing. Network Memory includes the definition of the non-TPN network and message queues used by both the CLM and the interface board.

Node

The individual components connected to the LCN are called nodes, and sometimes modules. Generally, nodes are logical and modules are physical. The node types are:

- US —Universal Station
- HG —Hiway Gateway
- AM —Application Module
- HM —History Module
- CM —Computing Module (DPS6 through the CG)
- CG —Computer Gateway

Node Configuration

The process of specifying the characteristics of a node. This means specifying the software parameters for a node, the units associated with the node, the card file layout and the type of processor card in the node. It also includes the process of assigning points to the slots defined in an LCN module or Hiway box.

Node Loading

This is the processing load on the control processor, resulting from the processing of control points. The processing load is measured in Processing Units (PU's). The processing load is dependent on the number of each point type and the frequency at which each point type is scanned and processed each second.

Other parameters like CPUFREE and PARSEC (Parameters per Second) in the PERFMENU displays also help calculate node loading.

Node Pair Number (np)

Notation used to represent an HM node pair number. In LCN Node Configuration, a node pair number is assigned to each HM.

Node Wide DDB (NWDDB)

A database (an HCI server) allowing communication between GUS Displays. The NWDDB differs from the DDB in that the NWDDB does not mimic the behavior of the US DDB.

Non-TPN Network

A non-TPN network and its associated devices are connected to the TPN through the CLM.

Normal Sequence

The portion of a sequence program that is executing under normal conditions.

Normal Value Status

The PV (Process Variable) value status is normal if all input values for the PV are normal, the PV is calculated as a real number and within range, and the PV source is AUTO.

Not a Number (NaN)

An indicator in a data point parameter that indicates that the parameter normally contains a number, but at present does not. At the Universal Station, a NaN data field appears as "- ----" (six dashes).

A floating point value, which is Not a Number.

Notification

Used interchangeably with "Event"

NT Application

An application built for execution as an NT executable (.exe)

NT Domain

An NT Domain consists of a group of computers and devices on a network that are administered as a unit with common rules and procedures. The NT domain provides an administrator a single point from which to administer user accounts, hard drives (shares), and network printers.

NT File System

The Microsoft Windows NT® file system (NTFS). It has security and recovery advantages.

NT Side

The Windows NT co-processor within a TDC 3000X TPN

NWDDDB Database

The concatenated RAM copy of all NWDDDB Database Input Files indicated in NWDDDB Server's log file.

NWDDDB Database Input File

A textual file that resides on the NTFS, and consists of OPC data items to be loaded into RAM at NWDDDB Server start-up. It is characterized by its ".dss" extension. This could be the Default NWDDDB Database, or any other user-provided file.

NWDDDB Log File

The list of all NWDDDB Database Input files to be loaded into RAM to form the concatenated NWDDDB Database. It is named "nwddb_server.log", and is installed at a well-known location.

O

Object Code

The lower-level code (machine code) that is the result of compiling a program written in a higher-level language (or code) called the source.

Object Dictionary (OD)

For Fieldbus:

Contains descriptive information about application objects accessed through VFDs (Virtual Field Devices). Each VFD maintains its own object dictionary that it shares with FMS (Fieldbus Message Specification). FMS uses information in the dictionary to encode and decode application data sent over the network.

Object Linking and Embedding (OLE)

A collection of object-oriented technologies based on a standard Component Object Model (COM). One OLE interface provides the Uniform Data Transfer of information between applications. Another OLE interface, Automation, enables cross-application macro programming.

Objects

For Fieldbus:

Entities within the User Layer, such as function blocks, a resource block, an alert object, trend objects, parameters, display lists, etc.

<object> Tag

Tag added to HTML language by Microsoft to include an ActiveX control in a Web page.

Offline

Activities performed that do not affect the operation of the plant or mill

Offline Application

OpenDDA Application that is not logically connected to the DCS for execution. These applications cannot be started by the DCS and cannot receive events from the DCS.

Off-Page Connector

An Off-Page Connector is an indication on a control drawing, where a connector leaves a page boundary. This indicator identifies where the connection can be found on another page of the drawing, in terms of page number and grid location.

Offsites Data Manager (ODM)

OM&S module

OLE Automation

Standard for application interfaces defined by Microsoft, providing a way to manipulate an application's objects from outside the application.

OLE (Object Linking and Embedding) automation is typically used to create applications that expose objects to programming tools and macro languages, create and manipulate one application's objects from another applications, or to create tools for accessing and manipulating objects.

For information on OLE Automation, refer to the **OLE Programmer's Reference**, Volume 2.

OLE for Process Control (OPC)

A set of industry standards developed by the OPC Foundation for to facilitate interoperability between applications within the Process Control Industry. These include automation/control applications, field systems/devices or business/office applications.

OPC (OLE (Object Linking and Embedding) for Process Control) specifies a standard interface to be used between two types of applications called OPC clients and OPC servers. An OPC server is an application, which collects data, generally directly from a physical device, and makes it accessible through the OPC interface. An OPC client requests and uses the data provided by an OPC Server. By having a standard interface OPC clients and servers written by different vendors can communicate.

An HCI server is also an OPC server.

OLE In-Process Handler

An in-process handler is a Dynamic Link Library (DLL) that runs in an OLE container's process space. It handles a limited set of the OLE objects services. There is default OLE handler provided by Microsoft that handles drawing and copying of an object without activating the local server. It is used primarily with local servers.

OLE In-Process Server

An in-process server is a Dynamic Link Library (DLL) that runs in an OLE container's process space. It handles all of the OLE objects' services.

OLE Local Server

A local server is a standalone executable that runs in its own process space. It handles all of the OLE objects' services.

On-Process

A mode where the system is on-line controlling the process. The opposite is off-process.

On-Process Intersection

The portion of the on-process functionality that is also required in the off-process personalities. Usually outwardly visible as a subset of standard displays such as System and Console Status.

Online

Activities performed that may affect the operation of the plant or mill

Online Application

OpenDDA Application that has been logically connected by use of the Install Application command line function to a DCS for execution. These applications are typically started by the DCS and may receive events from the DCS.

Online Reconfiguration

Changes to Network Configuration that are made with the configurator in the online mode. Such changes require that specified nodes be shutdown and reloaded with data that includes the revised network configuration file (NCF) with the current timestamp.

Open Data Definitions and Access (OpenDDA)

Open Data Definition and Access is a platform independent data abstraction layer that:

- provides a standard application programming interface
- assistance for developing and implementing applications with the following qualities:
- perform efficiently
- read and write varying amounts of heterogeneous data
- access one or more external data sources

Open Database Connectivity (ODBC)

Open Database Connectivity is a standard developed by Microsoft. It uses driver's specific to various types of databases to create a uniform way for programs to access these databases, regardless of database vendor.

OpenDDA Application

A Program written in FORTRAN or C/C++ that uses OpenDDA extensions to the language (EDB and EXEC DDA statements) to access data on a DCS using DCS independent syntax.

OpenDDA Application Directory

The directory where your OpenDDA application physically resides. This directory is usually (but not required to be) where the application was created, compiled, and linked. The files which support your application (that is, CIO file, and DEF file) are also contained in this directory. For on-line applications, this directory is the directory from which the Install Application was performed. For off-line applications, this directory is the current directory.

OpenDDA Status Mnemonics

Mnemonic names for constants used in an OpenDDA application when dealing with OpenDDA status values. The mnemonics are used exclusively throughout the OpenDDA user documentation whenever a reference is made to an OpenDDA status value.

OpenDDA Status Values

Values returned to an application by an OpenDDA Execution Statement. These values are used to indicate various degrees of success or failure.

Operating Group

Consists of values, status, and trends for up-to-8 data points, which are usually associated with a particular part of a process, assigned to a display for the convenience of monitoring or changing operating parameters and control modes.

Operating System (OS)

For computers, for example: NT, DOS, UNIX

Operator (OPER)

Usually, a process operator at a Universal Station (someone who doesn't have a supervisor's key nor an engineer's key).

OPER is the lowest key level from which parameters can be changed.

Operator Console

A set of one or more Universal Stations grouped together physically, and with Operator (or Universal) software usually loaded into each Universal Station. This can differ significantly from the way a "console" is defined during configuration of a system. See Logical Console.

Operator Interface Station (OIS)

US in ergonomic "Z" furniture

O

Optimum Replaceable Unit (ORU)

Optimum Replaceable Unit (ORU)

A replaceable hardware assembly that provides the best trade-off between its replacement cost and the cost of isolating faults within the unit.

Output Focus

A concept involving the automatic replacement (closure) of an existing application display by a new application display. When an existing display has output focus this means that it will be replaced by the next, newly invoked application display. In multiple-group workspaces, each round robin and manual select window group has its own output focus that determines which window in these groups is replaced when new displays are managed in these groups.

See Global Output Focus.

Output Focus Button

A button on the title bar of all managed application windows and placeholders (appears in the form of an ">" on Windows NT 4.0) that indicates whether or not a window currently has output focus status. You select this button to give output focus status to the window. If one window has output focus and you select another window in the window's group for output focus, the previous window's output focus button is automatically deselected.

Owning Node

The node on the LCN in which an entity resides. This node contains the external/internal entity ID-conversion capability, knows whether a parameter exists on the entity and, ultimately, contains the data for the entity.

Also known as the Data Owner Node.

P

Package

A collection of (usually software) pieces that are pulled together for distribution, sale, licensing, and installation. From “packaging” – the notion of pulling things together to make a single entity for distribution, sale and installation. Note: a collection of things can be referred to as a package as long as the packaging notion exists for distribution and sale even if the installation of the package components have not been coalesced into a single procedure, as is normally desired.

The term used in context may be taken to have any of a number of specific meanings, such as:

TPS Application Package A collection of the components of a TPS application that are packaged for distribution, sale, licensing, and installation.

TPS Component Package A collection of TPS components that are packaged for distribution, sale, licensing, and installation.

Page Break

In a control drawing, dashed lines that indicate page boundaries. Viewing page breaks is optional.

Parallel I/O

A Logic Manager/Logic Controller electrical interface designed to support the interconnection of multiple I/O racks with high-speed processing.

Parameter

Named database items that contains a value (real, integer, logical, string, etc.) or an array of values. The parameter values are accessible through the data point name and parameter name.

The values can be:

- input data_flow, temperature, pressure
- tuning parameters_gain, time constants, bias, recipe data
- output data_valve outputs, alarm states

The TPN/TPS bookset includes parameter reference dictionaries which list and define every parameter associated with specified system devices.

TPS Builder

A TPS Builder parameter has a name, type, value, and a set of attributes. Most TPS Builder parameters are also TPS Network parameters. Three types of parameters are distinguished:

- Identification parameters: unique name (same as tag name); description, type, revision number, revision date, author, etc.
- Standard control parameters are the loadable parameters that define a process control strategy
- Custom parameters are user-defined; they may be loadable to the system or database-only, as required

Parameter Configuration Form (PCF)

For TPS Builder/Fieldbus:

The TPS Builder dialog box from which parameters on a template/strategy can be configured.

Parameter Description File (PD)

The PD file is a text file that contains the parameter descriptions and the Manufacturer Enumeration descriptions.

Parameter Entry Display (PED)

The information on an entity being built by the Data Entity Builder that is displayed on the US screen. The information consists of multiple displays, called the PED set, and the status of the data in each display is available in the PED Set Status Display.

Parameter Group

A parameter group is a logical grouping of parameters. The group is visible as a labeled tab in a parameter configuration form.

All parameter configuration forms include Honeywell-assigned (system) parameter groups for configuring system parameters.

For example, alarm-related parameters are grouped together on an Alarming Display tab for applicable strategies.

Parameter List

A parameter list is a list of the names and types that the parameters of a TPS system data point can have.

At compile time, the compiler can refer to a parameter list for a description of the referenced data points. As such, the parameter list serves as a surrogate for the on-process database (because the database may not yet be fully described).

For TPS Builder:

In TPS Builder, parameter lists are represented as parameter groups that appear as tabbed pages of the parameter configuration form.

P

Parameter Reference Dictionary (PRD)

Parameter Reference Dictionary (PRD)

A compilation of the parameters defined in a particular Honeywell product or system, which includes the type, use, etc.

There are several PRDs in the Honeywell TPS bookset including the *HPM Parameter Reference Dictionary*.

Pascal

Structured programming language derived from Algol

Passive Property

A property that simply holds values. The value of a passive property may be used to influence behavior, but the act of writing to a passive property does not invoke an IOMap function.

Pathname

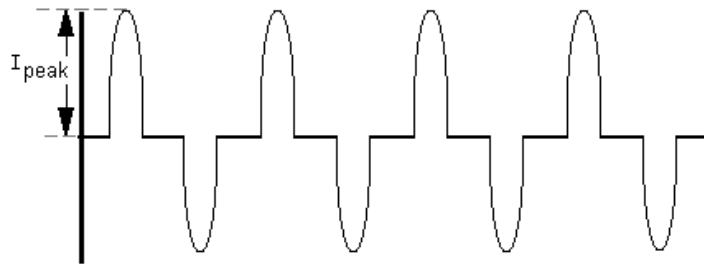
The description of source, volume or directory name and file name that locates a named file (for example, NET>VDIR>FILE.XX).

Pathname Catalog

An Area Data Base entity that defines the storage location of graphic displays stored on History Modules or removable media.

Peak Current

This is a measurement of the normal running current, but instead of an average or RMS value, the peak value is given. This is the highest value that the current reaches during a given half cycle. An illustration of the measurement of peak current is shown below. The waveform shown is typical for a load of the rectifier/capacitor-input type.



Peer Map

Peers are connected PM-family entities that are assigned to different nodes. Peer communication is achieved by a connection through a Universal Control Network (UCN).

TPS Builder can generate a connectivity map that includes peer connectivity information.

Peer Statistics Display

This display presents peer information for a project for which a connectivity map has been generated. Information includes a list of all nodes represented on the map, the number of inputs and outputs for the named node, the number of points externally referenced in CL, and the number of invalid peer connections.

P

Persistence of External References

Persistence of External References

The procedure of modifying the External Reference Components, saving the memory resident CIO to disk, reloading the CIO later, and using the External References, is known as persistence of External References. The dynamically selectable External References persist over time by writing them to the CIO on disk.

Personality

A specific software image in a node on the LCN. Nodes can typically have multiple personalities but they don't coexist. For example, the US has the Operator's Personality and the Universal Personality. Each requires the complete memory load to put the personality in operation.

Other nodes have an online personality and an offline personality such as History Modules (HM).

The 68020 processors (K2LCN or LCNP) and 68040 processors (K4LCN or LCNP4) also have personality files.

Phase

A sequence program consists of a series of phases. A phase accomplishes a major process function or marks a major process milestone. Examples are a charging phase or heat-up phase. Phase boundaries are key points of synchronization in a sequence program.

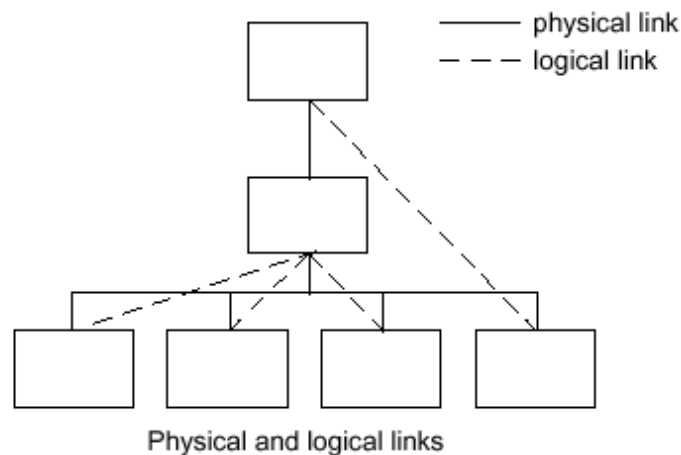
A phase is identified in the sequence program by the PHASE statement, and is divided into steps.

Physical Entity Control Module (PECM)

The PECM is a subset of the TPS Control Module (CM) subtype. The PECM represents a device on a Fieldbus link.

Physical Link

The physical link defines the interconnection of systems within the communication network. Multiple logical links may be realized across a physical link.



Pick

A Universal Station target that allows you to select operational alternatives or initiate an action.

Picture

See Schematic. **Picture Manager**

A Honeywell-provided software component providing enhanced GUS display call up performance on GUS nodes.

P

PIN Global Address

PIN Global Address

An optional method of PIN node identification that is not currently supported for Network Gateways (NG).

PIN Local Address

A two-digit number that uniquely identifies each NG on the PIN. Established by physical pinning on a communications board.

Pinning

A means to define in hardware the address of a node on a network

Pixel

The smallest display element on a CRT (a dot).

Placeholder

A blank window associated with a particular window specification, created by SafeView at runtime. It represents the actual application display window had one been managed according to that window specification. A placeholder represents a window specification or ViewPort when there is an actual application on display. Each ViewPort can be configured to either have a placeholder, or not have one. When no application is being managed by using a given window specification and the placeholder option is specified, the placeholder defined in the window's specifications is displayed. It is removed from display when a real application display is associated with the window specification. The placeholder can be used to manually select where the next application window associated with a manual select group will be displayed.

Plant Control Network (PCN)

See Process Control Network

Plant Information Network (PIN)

The Plant Intranet is the open network that exists throughout the plant. It is used by plant personnel for viewing process data, but is not guaranteed to have near 100% availability.

Plant Intranet

An open network that exists throughout a plant, is used by plant personnel for viewing process data, but is not guaranteed to have near 100% availability.

Platform

The term platform is used to reference the hardware and software computer system serving as an external data source. Platforms supported by OpenDDA include Honeywell's Application Module (AXM)

PM Family

The PM family consists of the Process Manager, Advanced Process Manager, and High-Performance Process Manager controllers.

Also referred to as xPMs.

Point

A TPS Process Network data point contains all of the information about a given process-related entity (for example, a regulatory control point) required for system configuration. Data points reside in a data owner, such as a NIM, AM, HG, or CG. As a rule, its TPS Builder representation is a strategy/template block.

Data points are the basis for process data acquisition and control. A data point is a named collection of data values called parameters. The parameters contain data related to the process and to the data acquisition and control functions.

Data points in your system are defined as you design the control strategy for your plant or process.

Point information is saved in:

- Checkpoint files – manually or periodically
- EB (Exception Build) files
- IDF (Intermediate Data File) files

Point Building

The process of defining the content of a data point.

Point Interface Table (PIT)

The Point Interface Table contains the relationships between CLM image points and their corresponding actual non-TPN network points, along with conversion and data acquisition information. Each image point in a PIT is represented by a PITE. Each PIT defines a set of points grouped together, based on both processing time and physical device address. Each PIT must be related to the NDT that defines the addressable network component or device where the set of points is located. A PIT can also optionally be a member in a PITG.

Point Interface Table Entry (PITE)

The PITE defines one image point within a PIT. The PITE consists of parameters that define the interface input and output requirements between the CLM image point and the non-TPN network point. Each PITE within a PIT must have the same processing cycle

rate and the same physical device address, although the specific point or register is unique to that PITE.

Point Interface Table Group (PITG)

A PITG optionally defines a set of PITs to be processed as a group. This allows same-time processing of PITs across different devices. A PITG represents the highest level entry in the overall point interface structure and, if used, must be defined prior to the definition of member PITs within the group. A PITG may not cross ports.

Point Name

A point name is a unique name identifying a data point (also called data point name, or tagname).

Each name is from one to eight or one to sixteen characters long and the first character must always be an alphanumeric character.

Examples: TIC100, FC_2000, PUMP200A

Point names reside in data owner nodes such as: AMs, HGs, NIMs, and CGs.

Point Process Special (PPS)

See Process Special on a Data Point.

Point Type

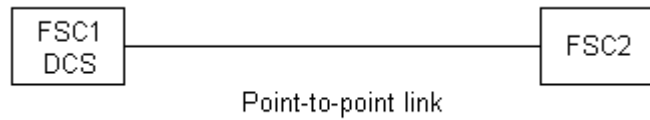
Data points that accomplish a specific type of function. Regulatory, Timer, Numeric, Switch, and Custom are examples of point types.

Point.parameters

See Parameters.

Point-to-Point

A point-to-point link is a physical link that interconnects two systems, with one system connected to each end of the link.



Policy Editor

A Windows NT tool used to establish a certain set of operating and configuration rules on a node.

It is a high level tool through which Registry values can be set. This determines how NT and the applications running beneath it behave.

Populate HLL

High Level Language (HLL) variables associated with the External Data Statement will be populated with values read from the external locations.

Port

A Universal Station target that is available for the user to enter a specified piece of data into the database.

Pounds per Square Inch Gauge (PSIG)

Above 1 atm

Power Factor (PF)

This is the relationship of the actual power usage to the apparent power usage. In reality, a given unit may use only 100 watts of power when averaged over a period of time (the actual power), but if the RMS current and the RMS voltage are multiplied together (the apparent power), a different higher result is obtained. For this example (100 watts actual power), let us say that the RMS current is 1.28 A and the RMS voltage is 120 Vac. The apparent power would be $1.28 \text{ A} \times 120 \text{ Vac} = 153.6 \text{ VA}$. The power factor for this example would be $100 \text{ watts} / 153.6 \text{ VA} = .651$ or a power factor of 65.1%.

Power On Hours (POH)

How long a unit has been powered

Power Supply Module (PSM)

Each electrical device has its own power supply, usually 24 V.

For Logic Manager:

Module supplies power to operate the processor. A battery compartment contains a lithium battery, which provides backup power to the memory and register modules.

Preemption Point

The point where suspension of a sequence program occurs to allow another program to run.

Preferred Access Device

A device on the Data Hiway that can request permission from the Hiway Traffic Director to access data in a process-connected box.

Prefetch

When a data point requires data from a data point in another module or gateway, that data is fetched (read and transferred) before the destination point is processed, hence the term **prefetch**.

Prefetch is used mainly in AMs before points are executed in each cycle.

Primary Data Point

A data point in a cascade-control strategy that supplies an input to a secondary data point. The secondary data point is the closer of the two to the process.

Primary Domain Controller (PDC)

Maintains the master Windows NT Systems Administration Management (SAM) database. The Primary Domain Controller tracks changes made to domain accounts and stores the information in the directory database.

Each Windows NT domain must have a single PDC.

Primary IOP

IOP B is the physical primary IOP. It is the IOP that is physically connected by cable to the “B” cable connector on a redundant FTA.

Primary Module

A Process Module data point that represents a collection of equipment that is, in turn, represented by a collection of data points. The Primary Module is assigned to a batch for the collection of batch history. Most data points have a parameter that can name a Primary Module, and all data points that name a specific Primary Module are assigned to it.

Also refers to the node that is controlling the process.

Primary Server

When using Redirection Manager, this term refers to the OPC server that communicates with client data sources, and is configured to be the Primary for an RDM. The same server could be configured as a Secondary for another RDM.

Printed Wiring Assembly (PWA)

For example, Plug-in Board

Prior Display

Pushing the Operator Keyboard button PRIOR DISP recalls the last display that was removed from the screen. You can reinvoke a prior display merely by selecting it and pressing the Invoke button on the Display History dialog box.

- Pressing Ctrl+Alt+P will also invoke the prior display.
- Only registered application displays can be invoked.

Process(es)

Software that executes autonomously in a single node

Process Alarm

An indication (audible or visual) that notifies a console operator of an abnormal condition in the process

Process Control Network (PCN)

The portion of the Ethernet network which connects process control nodes. It is typically isolated from the Plant Information Network (PIN) by a router for security and robustness purposes. Higher reliability implementations of the PCN can be called Industrial Strength Networks.

Process Data

Data that represents the state of the plant or mill, for example, temperatures and flows

P

Process Manager (PM)

Process Manager (PM)

A process-connected device on the UCN that provides regulatory control and sequence operations. The PM includes a powerful set of pre-built data acquisition and control algorithms.

Process Module Data Point

1. A platform for the execution of a sequence, or
2. A data point that represents a specific group of other data points that, in turn, represent a specific set of process equipment.

Either (1) or (2) or both may apply. If (1) applies, the Process Module data point is assigned to a sequence slot in an MC.

Process Module Group

A collection of up-to-8 Process Module Data Points that can be displayed on a single screen.

Process Module Operational State

The operational condition of the process module as controlled by the sequence, such as Off, Ready, Normal, Hold, Shutdown, Emergency, and Restart.

Process Monitor (PMX)

Software package running on 45000 Honeywell computers

Process Network

A communication network that links the LCN to those devices that are connected directly to the process sensors and actuators. The Data Hiway and UCN are process networks.

Process Operations Management System (POMS)

IBM OS/2-based software package

Process Optimization

The concept of the application of control hardware and software to maximize a process performance, usually for financial or economic value.

Process Special on a Data Point (PPS)

A condition that causes immediate processing of the data point. A process special may be requested by an operator, a CL block, or event-initiated processing, or any other mechanism that can store a value in the Process Special (PPS) parameter.

Process Unit

A portion of the process in which some specific type of operation takes place. For example, a boiler, a furnace, a distillation tower. A unit is any logical grouping of points, usually aligned with some subdivision of the process.

Process Variable (PV)

The value of some time-varying property in the process; for example, a temperature, flow, and pressure. PVs can also be calculated from two or more measured or calculated variables. One parameter of a data point.

Processing Cycle

AM (Application Module) data points can be assigned to a predefined processing cycle within their assigned processing interval.

Processing Unit (PU)

A weight given to control processor point types in Process Managers, Advanced Process Managers, and Logic Managers that is used to calculate an acceptable mix of data points for that device.

Product Authorization Key (PAK)

CM50N License Management Facility

Product Path

The selection of process modules used to make a batch.

Programmable Logic Controller Gateway (PLCG)

An LCN node that functions and appears to the software and the rest of the system, as if it is an HG connected to a Data Hiway with up to eight Data Hiway Ports. The PLCG is a fast, cost-effective interface to PLCs from several manufacturers.

There is also an enhanced version known as the EPLCG.

Project

Projects allow you to organize and present a related group of control strategies, to reflect a hierarchy maintained for the project. The standard hierarchies are the equipment hierarchy, control hierarchy and hardware hierarchy, but these can be changed for a project if required.

Projected Parameter

For Fieldbus:

TPS Network accessible parameter for which the IOP has been configured with binding information regarding access to the data corresponding to the parameter from the Ultimate Data Owner (UDO).

Project-Up Parameter

For TPS Builder:

A parameter is projected up when it is defined in a strategy/template contained in a module and it shows up on the module's list of parameters. By default, none of the parameters of contained strategies/templates are defined as project-up.

Propagate Down

A control attribute applying to modules (template and strategy). This attribute allows you to set a parameter value at the module level.

The value is then applied to all elements of the module (if that parameter is part of the element's configuration).

For example, for TPS system parameters, only NODETYP, NTWKNUM, and NODENUM exist for both node modules and the points they might contain; setting of these parameters is a primary use of propagate-down.

Property

Specific characteristics of a MapItem, IOMap, or IOMap Server. Examples of properties are:

- Name
 - Cross Reference
 - Test Value
 - Test Mode
 - External reference
 - (maps abstract names to concrete external references)
-

Publisher-Subscriber VCR

The Publisher-Subscriber relationship allows multiple endpoints. A single publisher may have multiple subscribers. The data is Buffer, Network initiated, and Unidirectional (BNU).

The data may be cyclic as requested by the LAS or acyclic as requested by a subscriber. The publisher has a DLCEP address but the subscriber has no data link address.

PV Source Selection

The ability to select the real, calculated, or substituted value for the process variable.

PV Tracking

Without PV tracking, when the point is in Manual mode, unintended control action can occur. To prevent this, optional PV tracking causes the setpoint to track the PV value when the point is in Manual mode, so that when the point is again in Auto mode, the initial SP value is the same as the PV.

Q

Quality Engineering Graphics (QEG)

Part of EH800

Query

An inquiry concerning the system database.

Queued, User-triggered, Bi-directional (QUB)

For Fieldbus:

A type of VCR (Client, Server, and Peer)

Queued, User-triggered, Uni-directional (QUU)

For Fieldbus:

A type of VCR (Source and Sink)

Q

Queued, User-triggered, Uni-directional (QUU)

R

Rapid Displacement Heating (RDH)

Batch digester

Rate Of Change (ROC)

Used in alarms if the PV of a point rate of change is above the desired value, that is, changing too fast in either the positive or negative direction.

Read

To request and receive data from a data point parameter. Sometimes fetch is also used in this sense.

Read Data

All external data associated with the statement will be read in preparation for populating the High Level Language (HLL) variables.

Real Time Multi-Programming Operating System (RTMOS)

Operating system for the 45000 computers with PMX or PMC software

Recipe

All steps and processing parameters (temperatures, ingredients, times, and so on) required to make a batch of a given product.

Reconfigure

See Online Reconfiguration.

Reconnect

To physically reconnect a section of coaxial cable to a T-connector or to logically reconnect multiple nodes so that they communicate, as in LCN Reconnect.

Reconstitute

A Data Entity Builder command that reads an entity from the module in which it resides, converts it to PED (Parameter Entry Display) form, and displays it in the current PED.

The back-building operation—reconstituting an entity from its owning node or area database file to the Data Entity Builder for display.

Red Green Blue (RGB)

Refers to three lead coax CRT color signals

Redirection Manager Windows Management Instrumentation (RDM WMI) Client

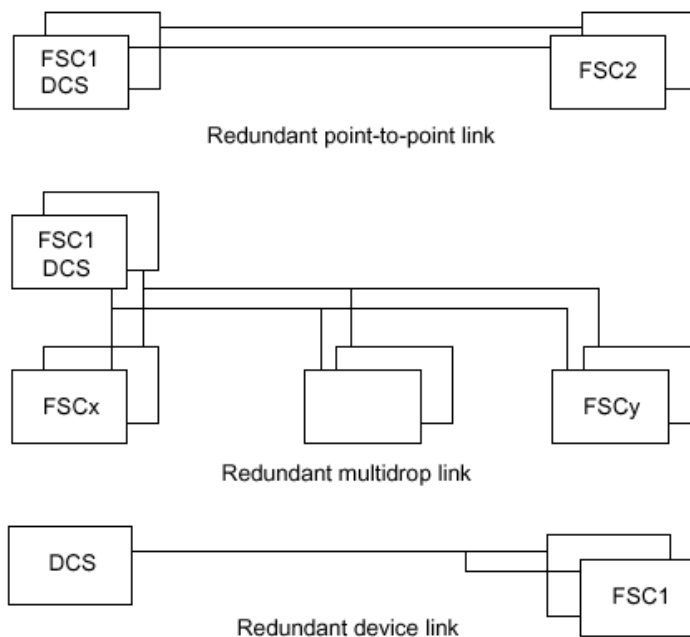
A subscribed component of the WMI notification requesting notice of changes in the status of Primary and Secondary servers. The RDM WMI client then notifies the Redirection server of changes so that appropriate action such as switching to the alternate server can be taken.

Redundant

Two nodes of the same type (one primary and one secondary) form a pair - the secondary backing up the other. For example: NIMs, HGs, and AMs.

Redundant Link

A redundant link is a logical link, which is connected to both Central Parts of a redundant FSC system. For FSC-FSC communication, a redundant link is realized through two independent physical links, each connected to a single Central Part. For communication with non-FSC systems, a redundant link is realized through a single physical link which is connected to both Central Parts.



Registered Application Display

See Workspace Client.

Registered Client

An Application or Application Component that is known to the TPS system

Regular Expression

A technique for describing lexical string patterns; a string-expression that can contain wildcard characters, so that it can match a range of actual strings. Example: the regular expression “Unit1?*” would match the string “Unit1Schemes” as well as “Unit1 Overviews.” Regular expressions allow match expressions to specify a range of values without enumerating each value.

Regulatory Control

Regulatory control uses a predefined set of control and auxiliary algorithms to maintain a continuous process variable at a desired value by manipulating an output. Regulatory data points in the HG and the AM accomplish regulatory control.

Regulatory Point

A point or loop involved in regulatory control.

Relationship Browser

For TPS Builder:

This dialog is available from the Tools menu. It enables you to evaluate the associations between several objects within TPS Builder. You can determine relationships between and view details of projects, libraries, templates, strategies, views, and attached documents.

Release Directory

A directory structure used to store the DD (Device Description) and related files. Each manufacturer has a directory name that is the register manufacturer id number. Under the manufacturer id directory is a directory for each device type that the manufacturer produces.

Remap CIO

Remap CIO is the functionality which allows an OpenDDA application to dynamically associate the HLL variables to a different group of external data by reading multiple CIO files into memory.

Remote

A coaxial segment that is connected to the central segment only by a Segment single LCN Extender Set

Remote Access Service (RAS)

NT web browser feature

Remote Control

The Carbon Copy 32 utility that allows a Guest user to control a Host PC. You can view the remote PC screen in a window on your desktop, or display the remote window full screen.

Remote IKB Client

Functional package that provides IKB/OEM functions at the GUS Client.

See Integrated Keyboard

Remote LCN

See Remote System.

Remote Node

A node on a network other than the AXM node that is used to obtain access to external data.

Remote Object Dictionary (ROD)

The ROD is the host base object dictionary into which a DD Binary is loaded.

Remote Step Processor

For Application Director:

An agent that executes steps of an Application Director application in a TPS node that does not have an Execution Environment but is under the direction of an Execution Environment from another TPS node.

Remote System

Nodes on the local LCN are able to exchange information with systems not attached to the local LCN. A remote system can be another Honeywell LCN network or another vendor's network and/or computer (future implementation).

For example, a Network Gateway (NG) or a TPN Bridge can be used to transfer information between the local LCN and the remote LCN.

Remote User LCN Access (RULA)

An optional software function with appropriate communication hardware that enable users at a remotely located Universal Station to see the displays at another ("Local") Universal Station, and to be able to perform certain designated functions as though they were present at the "Local" station.

Removable Media

Cartridges and floppy diskettes. Such a medium can be removed from its drive and replaced by another medium of the same type.

NOTE: For R500 and beyond, cartridges are the only supported removable media.

Reports

LCN

There are Area Data Base reports available on the LCN to access point information.

TPS Builder

Reports make information regarding TPS Builder projects or libraries, as well as overall statistics and other details, available for viewing, printing, and other activities.

Reports are formatted for printing, and include borders, column headings, and identification blocks typical to engineering documentation. They are generated by Seagate Crystal Reports software, the runtime version of which is provided with TPS Builder.

Request Identification (RID)

Customer complaint log

Requirements

Testable capability of the control system

Reserved Entity

A system-named entity reserved by the system (and therefore unavailable to users) for use by area and process network configuration items. All Reserved Entity names begin with \$.

For a complete list of Reserved Entities see Table 9-1 in the *Data Entity Builder Manual*.

Residence

The box, module, or gateway where a data point resides.

Resource Block

For Fieldbus:

The tagged database part of the VFD (Virtual Field Device). It is also a named block consisting of contained parameters, which provide an interface to common resource capabilities or information.

Responsible NG

The NG selected as the primary communications path between the local system and a remote system connected to the same PIN.

Remote GUS

A Honeywell product enabling a non-GUS node to connect to a GUS node acting as a server, in order to provide GUS functionality at the non-GUS node. GUS servers are limited in the number of clients supported at a given time. Note that even though Remote GUS is intended to run on a desktop (non-LCN connected) computer, it's GUS Mode is considered different then when executing on a desktop but not using Remote GUS.

Root Mean Square (RMS)

Root Mean Square is the square root of the mean square of the variable values taken throughout one cycle.

Round Robin Group

A kind of window group in which window specifications are applied (that is, new application displays are presented) in a least recently used fashion, for application windows that match the group's match expression. This typically results in new application windows replacing old ones, in a least-recent manner. This group can be thought of as a round robin style window group, if one positions the windows in a clockwise fashion.

Router

Divide networks logically instead of physically, similar to switches in that they can filter out network traffic.

For example, an IP router can divide a network into various subnets so that only traffic destined for a particular IP address can pass between segments.

Runtime

The period of time during which a program is executing, as opposed to compile time or load time.

Runtime Mode

One of two standard modes in which ActiveX controls execute. Runtime mode indicates that the ActiveX control is expected to be “operational”. Contrast with build-time mode (also referred to as design mode) in which the ActiveX control behaves as appropriate for configuration/build/design activities.

R
Runtime Mode

S

Safety Manager (SM)

A process-connected device on the UCN that uses three redundant modules to provide emergency shutdown functions through digital/analog I/O points. It can share data with other UCN nodes but isolates process control operation from safety or emergency shutdown functions.

SafeView

Honeywell system software used for windows management.

SafeView can be configured so that selected views of the TPS system are displayed on the screen. For example, Native Window can have the property of never being covered by another display. See *SafeView User's Guide*, GU27.

Sales Opportunity Activity Report (SOAR)

Forecasting system to drive factory inventory

Sample

Input Multiplexer

The data of a multiple input is read in parts. One part is a sample.

Output Multiplexer

The data of a multiple output is written in parts. One part is a sample.

Scan Time

Input Multiplexer

The time in which all multiple inputs of a multiplexer are read.

Output Multiplexer

The time in which all multiple outputs of a multiplexer are written.

Scanner Application Module (SAM)

Part of the TPS Paper Machine Automation System and executes the scanning and sensing features used in paper-making process activities.

Schematic

A schematic (custom display) is a user-defined graphic and textual representations of the process or any other available information for the console operator on the TPN system.

Also referred to as a display, graphic, or picture.

Scientific Apparatus Manufacturers Association (SAMA)

Logic diagrams

Scope Frame

Large left-side frame on the TPS System Status Display showing tree of Domain Name-Node- Component on the TPS Network.

Scope of Responsibility

An operator's area of Process Control Responsibility. In TPN, the Scope of Responsibility (SOR) is based on the TPN concept of an Area.

Script

A block of text containing GUS Basic. A script consists of zero or more functions, subroutines, and data declarations.

Secondary Data Point

A data point in a cascade-control strategy that receives at least one input from a primary data point. The secondary data point is the closer of the two to the process.

Secondary IOP

The secondary IOP is the logical secondary IOP. It can be either IOP A or IOP B that is not controlling the FTA's field terminals or providing the process variable.

Secondary Server

When using Redirection Manager, this term refers to an OPC server that can provide the same functionality as the Primary server and is configured as the backup server for the Primary server. This server can be configured as a Primary server for another RDM.

Segment

For Fieldbus:

A fieldbus sub-network in which any of the connected devices may communicate directly, with no intervening relay or bridge, also known as a Link.

Segment Control Module (SECM)

A subset of the TPS Control Module (CM) point type. A SECM point represents a single fieldbus link associated with a single FB-IOP (Fieldbus IOP).

Select Area, Load, Install, Change Area (SLIC)

To alter an Area Database

Selection Outputs

Input Multiplexer

The status of the selection outputs determines which sample of the data will be read. Only one of the selection outputs can be active at a time.

- **Active Time:** The time which the selection output should be active before the data of the multiplexer bus is valid.
- **Data Hold Time:** The time in which the data stabilizes on the multiplexer data bus.
- **Active Status:** Selection of the sample can be either active high or active low. On = active high, Off = active low.
- **Multiplexer data bus status:** The status of the multiplexer data specifies if the data is either represented normally or inverted ('normal' or 'invert'). The data is inverted after the data is read from the multiplexer data bus if the status is 'invert'.

Output Multiplexer

The status of the selection outputs determines which sample of the data will be written. Only one of the selection outputs can be active at a time.

- **Active Time:** The time during which the selection output must be active.
- **Data Hold Time:** The time during which the data on the multiplexer data bus must stay active after the selection output is deactivated.
- **Data Setup Time:** The time during which the data on the multiplexer bus has the correct data status before the selection output will be activated.
- **Active Status:** Selection of the sample can be either active high or active low. On = active high, Off = active low.
- **Multiplexer data bus status:** The status of the multiplexer data specifies if the data is either represented normally or inverted ('normal' or 'invert'). The data is inverted before it is written on the multiplexer data bus if the status is 'invert'.

Self-defining Enumeration

A parameter value that consists of an enumeration that is defined by the user. These self-defining enumerations are the character strings that appear on displays, reports, and logs to represent the state of flag, digital, and logical variables.

For example, for a digital input point, a PV value of 1 might represent the running state of a motor, so the self-defined enumeration for that value might be Running. Self-defined enumerations are configured by the user in the STATE1 and STATE2 parameters of applicable data points.

Selling Support Automation (SSA)

New OMS

Sequence

For Application Director:

A user-written program which controls the order of execution of steps within an Application Director application.

Sequence Execution

The process of interpreting and carrying out the instructions contained in a particular sequence program's lines of Control Language code.

Sequence Execution Mode

(Abbreviated to MODE on the Process Module displays.) Specifies the degree of operator interaction required to maintain the continuous, normal execution of a particular sequence program. In automatic and semiautomatic modes, operator interaction is required only if programmed in. In single-step mode, generally used for debugging programs, the operator must reinitiate execution at the beginning of each step of the program.

S

Sequence Execution State

Sequence Execution State

(Abbreviated to EXEC in the Process Module displays.) The status of the execution process for a particular sequence program. Some states indicate whether a program has been loaded into the controller (NOT LOADED, LOADING, LOADED). Others deal with the status while the program executes (RUN, PAUSE, FAIL, ERROR). The END state indicates that execution of the program is complete.

Sequence Program

A program for discontinuous control, bound to a Process Module data point, written in CL, and consisting of four major parts: Data Declarations, normal sequence, (made up of one or more phases), abnormal condition handlers (optional), and user-defined subroutines (optional).

Sequence Slot

A block of Process Manager/Advanced Process Manager, or Multifunction Controller memory locations into which sequence programs written in Control Language can be loaded and executed. Each is represented by a Process Module data point.

Serial Device Interface (SDI)

An I/O processor in the Process Manager and Advanced Process Manager that provides an interface for devices that use EIA-232 or EIA-485 serial communications.

Serial Interface I/O Processor (SI IOP)

An I/O processor in the Advanced Process Manager that provides a communications interface to Modbus compatible subsystems.

Serial I/O

A Logic Manager/Logic Controller electrical interface designed to support the interconnection of multiple serial I/O racks.

Serial I/O Module (SIOM)

This Logic Manager/Logic Controller module provides the serial I/O rack interface to the processor rack's Serial Link Module.

Serial Link Module (SLM)

This Logic Manager/Logic Controller module provides the processor rack interface to the serial I/O racks' Serial I/O Modules.

Server

The provider of services requested by a client

A process providing services to other cooperating processes

Service Access Point (SAP)

Application software co. teamed w/Honeywell Application software co. teamed w/Honeywell. R/3 product is an ERP

Setpoint (SP)

Setpoint (SP) is the desired value of a process variable. Setpoint is an analog point parameter, and the value is entered by:

- an operator
- another point in a cascade loop
- or a program

The setpoint can be changed any number of times during a single process. The setpoint is represented in engineering units.

Site Server

The enterprise server from which local (desktop) copies of the WebGUS product components may be downloaded.

Slot Allocation

In UCN node configuration, this function enables you to allocate the number of normal and fast slots for each type of control point in the node being defined.

TPS Builder calculates and displays the processor and memory loads as you enter the information in the Control Processor Slot Allocation dialog. For HPM nodes, TPS Builder also calculates the number of I/O link units required, which you can review and change if desired.

Smart Transmitter

XPM Smartline Transmitter subsystem – provides Read/Write communication between the system (STI IOP) and the transmitter.

Magnew 3000 – Flow measurement

ST3000 - Pressure measurement

STT3000 - Temperature measurement

Smart Transmitter Interface (STI)

An I/O processor in the Process Manager and Advanced Process Manager that provides an interface to Honeywell's ST 3000 line of transmitters.

Snap-in

Vendor supplied software that integrates with MMC to supply customizable views of administrative interfaces.

Source Code

A program written in a higher-level language (or code) that is the **source** for a compile operation, resulting in the object code that executes in a processor.

Space Systems Division (SSD)

Honeywell Space Systems Division

Specialization

This is the process of creating a template from another template and adding further definition to the newly created template. The new template inherits the entire definition of its source. For example, a motor control template may be a specialized definition of a digital composite template.

An association is maintained between two templates when one is specialized from the other. Many elements of the inherited definition are template defining and cannot be deleted from the specialized template. Some elements can be modified within the constraints set by the template-defining attributes. Also, changes to template-defining elements on a source template are propagated to its specialized templates.

Specific

A term applied to applications (“specific application”) that indicates that the application being discussed applies to a specific set of equipment or product.

ST 3000™

Honeywell’s line of digitally integrated “smart transmitters.” Also called **Smartline 3000**.

Standard Displays

Displays/schematics provided by Honeywell with a system at no additional cost.

Statement

One CL (Control Language) or SOPL (Sequence Oriented Procedural Language) instruction (line of code). A statement performs an elementary action.

Status Frame

Large right-side frame on the TPS System Status Display showing additional information about the entity selected in the Scope Frame.

Step

For Application Director:

A user-written or third party-supplied software executable that is part of an Application Director application.

For CL:

A milestone in a Phase that executes a minor process function such as opening a valve with associated checks and verification, or measuring a temperature. Steps are recognizable on Universal Station displays. They are identified in a sequence program by the CL STEP statement.

Step Processor

For Application Director:

Whether local or remote, the processing resource for an application's Steps, which are program executables.

Store

To save data into memory—to write to memory/disk.

To write a new value into a parameter.

Strategy

A TPS Builder strategy is the configuration information for some element of the TPS system (or another system) intended for loading to the system. A strategy contains loadable data, such as point names, parameter values, and slot assignments.

Typical TPS system strategies include system and point configuration data.

Strategy Browser

This TPS Builder dialog allows you to browse the TPS Builder strategy hierarchy and list of parameters associated with the strategies. The Strategy Browser is accessible from all parameter configuration forms in which data flow connection parameters exist, and it is also available in the CL Editor, for identifying strategies that you need to reference in a CL program.

STT 3000™

Temperature Transmitter in the Honeywell Smartline transmitter family.

Subsystem

A hardware or software component of TPS system, for example, GUS, NIM, HPM, Control Builder

Subsystem Options

In several TPS dialogs, a function called “subsystem options” allows you to narrow the TPS Builder activity by choosing a category of entities within your project. For example, you can forward build all entities in a project, choose only TPS Process Network or Data Hiway entities, or choose entities for a specific node or box, using subsystem options.

You can use this function in checking, forward building, and creating an entity list.

Suffix

For TPN:

The two letter extension of a file name that identifies the file type (for example, .EB, .EC, .DB). A period separates the suffix from the file name.

Supervisor

An individual in an operations center who oversees two or more process operators. One who has a key that can be inserted in the Universal Station's key switch to gain access to supervisor-accessible parameters.

Supervisory Sequence-Program

A sequence program that can initiate, monitor, and control other sequence programs that work together to make a batch.

Support Protocol Data Unit (SPDU)

For Fieldbus:

Used to support the full DL-protocol.

Symbol File

A file produced by the FF Tokenizer that contains symbolic names that were used in the DDL file for the Function Block Application Process of the device.

System

A computer or collection of computers with which a Network Gateway is able to exchange files and process control data. The generic term system is used in place of the proprietary term LCN to describe the remote systems with which the Network Gateway interfaces. This was done to provide for future phases of the NG development that are intended to allow it to operate in a multi-vendor environment where foreign networks and computing systems are supported.

System Control Module

A functional extension in a Logic Manager/Logic Controller of the Processor Module that coordinates the interaction of all the modules in the processor.

System Description

Consists of 20 characters describing a node (system) on a PIN.

System Event Server

An OPC server that exposes Windows events as OPC event so that the events can be integrated into the Experion PKS alarm and event subsystem.

System Environment

A subsystem that provides transport and high-level parameter access throughout the distributed system.

System Environment Parameter Access

That part of the system environment that provides the high-level parameter access.

System Infrastructure

Base support software that provides scheduling, management, and communication between TPS system software components. It includes the HCI.

System Maintenance Control Center (SMCC)

The Universal Station's Maintenance Software. The SMCC is an overlay to the Engineering Personality or to the engineering portion of the Universal Personality.

System Management Information Base (SMIB)

For Fieldbus:

Contains a set of variables that are managed by the System Management entity of a device. The data includes function block start time, a list of the VFDs within the device, timing data; features supported, and address assignment data.

System Management & Utilities

The System Management and Utilities provides for security, configuration, installation and status monitoring. It encompasses a huge variety of system internal tasks. It contains third-party toolkits (for example, backup, disk management, and event viewer utilities) to solve individual systems-management problems or cover a range for management of the network.

System Performance Server

An OPC server that can expose Windows Management Instrumentation data as OPC data items. This allows users to display, archive and analyze Windows system information within the DCS.

System Status Display

The System Status Display is used to view system and managed component status from all TPS nodes on the network. The status of each component is retrieved for display from the System Status Monitor repository.

System Title

Consists of one or two alphanumeric characters that uniquely identifies a node (system) on a PIN. At least one of the characters must be an alpha character (nonblank, nonnumeric).

T

Tag

Designation applied to the name given to a user data point. Consists of either up to 8 or up to 16 characters, depending on the system option selected, identifies the data point on displays, logs and reports, and is used by the system to identify the data point and to access data in it.

An entity that is being collected and historized. Equivalent to a parameter on the source system, for example, FC100.PV.

The identifiers for HTML document elements, like tables, forms, images, lists, objects, etc.

Tag Name

The name of a data point. Configured by the user in data point parameter NAME. Consists of up to eight characters, identifies the data point on displays, logs and reports, and is used by the system to identify the data point and to access data in it.

Target

A designated area on a Universal Station display that is available for the user to make a selection (pick) or to enter data (port).

Rectangular areas in the picture that initiate some specified action when selected by the operator. The action is defined by using an **action procedure** (actors).

TDC 3000

The primary control system offered by Honeywell prior to release 410. This includes the TPN, UCN, the Data Hiway, and connected field devices. It is now referred to as the TPS Network or TPN.

TDC Application

An application built for execution on a TDC System.

T

Technical Assistance Center (TAC)

Technical Assistance Center (TAC)

Honeywell's assistance center for customer telephone inquiries

Technician Utility

A PC-based tool that connects to the Auxiliary Port. This being the Configuration of the First VIOPs File and Card numbers (Base DPA), Firmware Download and Analog Module Calibration.

Template

A reusable configuration stored in a library. It is intended as the basis for producing strategies and is not intended for loading to a system. Typically it does not contain point names, slot assignments, or other application-specific data.

Template Definition File (TDF)

A TDF provides a complete definition for one type of CLM interface structure.

Each template is actually a separate file resident on the LCN (file.zt). The set of template files that encompasses the different types of CLM application interface structures is utilized by the CLM Configuration Builder. This interface structure build process determines format, constants for the entry type, and required user inputs with corresponding validation criteria.

Template-defining

Template-defining parameters are those parameters whose values can *only* be changed on the template where they are set as template defining.

Once these are set, all strategies created from the template, or new templates specialized from it, will include the template-defined parameter values.

Text Editor

For GUS:

A simple MDI Unicode text-based editor program (TextEdit) used to create and edit workspace configuration files. It includes an option to verify correct workspace syntax.

Although the majority of the functionality provided by the Text Editor has been duplicated by the new SafeView Graphical Workspace Editor (GWE) application, the Text Editor has been retained to provide the functionality that is not provided by the GWE.

For LCN:

The LCN text editor is used to build ASCII files such as EB and EC files.

Thermocouple (TC T/C)

A device consisting of two dissimilar metals that when heated produce a dc voltage. Used to measure temperature.

Timestamp

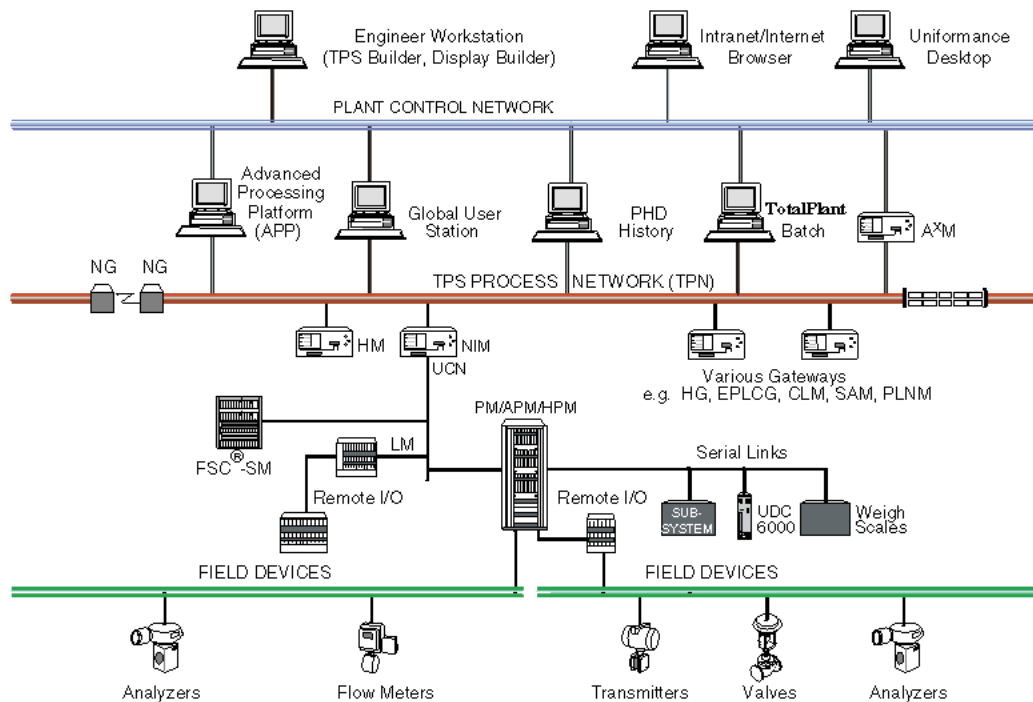
As a verb, the act of putting the current time together with an event. As a noun, the time value held with an event

Tool

A software program that helps a user does a job. Contrasted with an application that has an autonomous job to do.

Topology

A drawing showing how the nodes are laid out in the system and where they are located.



Total Distributed Control (TDC)

Total Distributed Control - the Honeywell design philosophy of full distribution of architecture, full integration of functions, and distributed risk, allowing for continuous expansion and capability.

TotalPlant™

Honeywell's program that commits to working with our customers to unify their field, process, production, and business management systems, using an open solutions approach.

TotalPlant Solution (TPS)

Beginning with R500 and with the introduction of GUS, TDC 3000 X evolved to become the **TotalPlant** Solution (TPS system for short). This name applies to any R500 or greater system, regardless of the presence of a TPS component.

Touch Screen

An option on the Universal Station that allows selections to be made by touching an area on the display with a finger or other cylindrical object.

TPD Application

An application that uses TPD support and execution software

TPN Bridge

TPN Bridge is a device that transparently breaks up large TPN systems into segments for a lower overall amount of traffic on a given TPN segment while speeding up the token circulation.

TPS Application

A collection of one or more of the above, designed to serve some customer need, and designed to run on a TPS system.

TPS Component

A software entity that is part of a TPS product.

TPS Domain

The namespace of a TPS system and all physical TPS nodes defined within that namespace. It is defined within an NT domain and uses the NT's domain's names for physical nodes, user Ids, and user groups for security checking.

TPS Network (TPN)

TotalPlant Solutions Network: The name for the Honeywell TDC 3000 distributed LCN control System. This includes the AM, NIM, HG, CG, HM, US, and other data owner nodes. The TPS Network encompasses the UCN devices.

TPN is synonymous with LCN.

TPS Node

This is a general generic term that collectively refers to an Intel-based workstation or server running Windows NT and having an integrated LCNP PCI card.

A node on which TPS runs and is part of a TPS domain. It can refer to a GUS node, APP, PHD, and/or other types of TPS nodes.

TPS Object

A named entity within an HCI Server Component, which contains data that can be fetched or stored from elsewhere in TPS.

TPS Server

An Experion Server TPS (ESVT)

TPS Station

An Experion Station – TPS (ES-T)

TPS System

TotalPlant Solution (TPS) system refers to the entire spectrum of Honeywell offerings, from the Global User Station to all existing TDC 3000^x equipment.

Trackball

An optional Universal Station device that allows quick and easy positioning of the cursor on the display screen.

Transceiver

CLCN A/B or LCNE or LCNFL boards which provide the electronic interface to the LCN coaxial node.

Transducer Block (TB)

For Fieldbus:

A software interface to sensors such as temperature, pressure, flow, and actuators that provide digital and analog output.

Transducer Blocks may be of two main types:

- Fieldbus Standard
- Manufacturer Specific

Manufacturer-specific blocks may be derived from a Standard block or a generic block.

Transducer Block templates have Fieldbus Link view.

Trend

A set of data usually shown as a graph, which allows you to see how a parameter's value relates to previous values of the same parameter.

True RMS current

The acronym RMS stands for Root Mean Square. The definition of Root Mean Square is the square root of the mean square of the variable values taken throughout one cycle. The formula for Root Mean Square current would be:

$$I_{\text{RMS}} = \sqrt{\frac{1}{T} * \int_0^T i^2 dt}$$

T = the time period of one cycle
i = the instantaneous value of the current

Typical values

The current values shown as typical (Typ) in the tables in this appendix are values that you would measure from typical units running at 120 Vac RMS. The values given are, in fact, values measured from a sample of units.

U

UCT

Universal Control Time

Uncertain (Value Status)

The PV source is MANual or SUBstituted, or if it is Auto, some input to the PV calculation is uncertain.

Uniform Resource Locator (URL)

Internet pathname/address

Example: <http://www.honeywell.com/>

Unit

See Process Unit.

Universal Control Network (UCN)

Honeywell's coaxial-based, dual communications network based on ISO open system communication standards that interconnects specified Honeywell process-connected devices with one or more Network Interface Modules.

The network that interconnects Process Managers (PM) and other UCN devices (APMs, HPMs, LMs). The Network Interface Module is an interface module between a UCN and the LCN.

Universal Control Network Extender (UCNE)

The Universal Control Network Extender (UCNE) is a fiber optic link that joins two UCN cable segments.

U

Universal Coordinated Time (UCT)

Universal Coordinated Time (UCT)

Same as Greenwich Mean Time (GMT)

Universal Personality

A Universal Station software package that combines the functions of the Operator Personality and Engineering Personality so that both are available for immediate use at a single US without reloading.

Universal Station (US)

The TPN system's principal human interface, the Universal Station (US) is a node on the LCN that provides the window to the process. It is designed to serve primarily in an Operator Console in a control room environment. It includes a color monitor, Operator's Keyboard, optional touchscreen, optional Engineer's Keyboard, optional Removable Media Drives (cartridge or zip drives), and optional trend pen recorders.

Universal StationX (UXS)

A Universal Station with an imbedded coprocessor containing workstation technology that includes a UNIX operating system and MIT's X Window System software.

Universal Work Station (UWS)

Compact alternate human interface, designed for location in an office setting for use by process engineers, process supervisors, and maintenance technicians.

Upload

Upload is the transmission of CM point, Fieldbus link, and Fieldbus device data from the TPS Network with Fieldbus to TPS Builder.

Upper Level Processor (ULP)

Host computer; part of the Computing Module. A computer interconnected to the TPN through a Plant Network Module or Computer Gateway.

User

Includes operators, engineers and other plant personnel who use the TPS system

User Access Panel (UAP)

Part of PCNM; window-based

User Request Service (URS)

Part of PCNM Server

U

User Request Service (URS)

V

Value

The content of every parameter of a data point, which may consist of real numbers, logical states, character strings, or enumerations.

Values provide live numeric data or status conditions on the screen. A picture editor *value* command is used to define a value. The value is entered as an expression. See the definition of Expression.

The value(s) stored in a parameter. Can be a real number, a logical value, a data-point (tag) name, a character string, or an enumeration.

Value File

For Fieldbus:

A text file, in Fieldbus Foundation (FF) Common File Format, that contains the configuration of a particular device, or device type.

A Value file can be used to update a Device template in a library, or a Device strategy in a Project.

Value Status

A parameter whose value represents the quality of the PV (Process Variable) parameter in a data point. The quality is specified as Normal, Uncertain, or Bad.

Variable

A name form consisting of a point name and a parameter name (for example, A100.PV)

Variant

Variants are used to choose one of several objects for presentation in the picture. The conditional behavior of a variant is programmed by using a simple textual language. The primary component of the language is an 'if' statement in which the limbs (or branches) of the **if** specify subpictures or text to be displayed depending on the conditional value in the **if** statement.

VB

Visual Basic

Very Large-Scale Integration (VLSI)

>100,000 transistors

VIEW Parameters

For Fieldbus:

Parameter sets supported by a Field Device to support efficient data access over the fieldbus network. Four types of VIEW parameters are defined by the Fieldbus Foundation.

ViewPort

A user-visible rendering of a window specification. A paradigm essential to understanding and communicating several of the features and concepts existing in SafeView. A ViewPort is a sticky window in which multiple applications can be consecutively displayed. It lends a sense of continuity, different than standard Windows, to display windows. The operator thinks of a ViewPort as a kind of static window that can exist for a very long time. Over a period of time, it can hold a number of different displays (one at a time). If configured appropriately, a ViewPort can be repositioned and resized; subsequent applications presented in this ViewPort would then be presented in this new position and in the new size.

Virtual Address Extension (VAX)

A computer made by Digital Equipment Corporation based on 32-bit architecture.

- The Computer Gateway (CG) of an LCN is used to communicate with a VAX computer.

Virtual Communications Relationship (VCR)

For Fieldbus:

Communication channel between devices or remote application processes. A VCR contains FMS, FAS, and DL-mapped information.

The major types of VCRs are:

- Publisher-Subscriber
- Client-Server
- Report Distribution-Receiving

Virtual Device

The set of IOMaps, MapItems, and the value of their properties are the state of the IOMap virtual device. An application, using an IOMap Server and creating IOMaps or loading IOMaps from IOMap map files, creates an IOMap virtual device. The application then performs I/O against this virtual device, using IOMap services by way of OPC.

Virtual Field Device (VFD)

For Fieldbus:

Defines an interoperable network view of a device application. Each Virtual Field Device (VFD) specifies the communications characteristics of its device application and the Fieldbus Message Specification (FMS) services used to access it.

The management or FBAP (Function Block Application Process) section of a device that is addressable by a VFD tag.

Visitor Device

For Fieldbus:

Temporary Device attached to the fieldbus network (for example, a hand-held configuration tool).

Volume

For History Module:

A logical named subdivision of a History Module's memory capacity, analogous to a drawer in a file cabinet. Volume names and sizes are defined during HM configuration.

System-reserved volume names begin with "!" or "&".

For Removable Media:

The system designation for each formatted cartridge disk or floppy diskette. The name by which a cartridge, zip, or a section of an HM is known.

See Directory.

Volume Name

A four-character, alphanumeric volume designator.

W

Watts

This is the integral of the instantaneous voltage multiplied by the instantaneous current over a half cycle. It is the actual power usage referred to in the definition of power factor above.

WebGUS

A product enabling;

- The GUS Run-time to be embedded as a component of a web page.
- A data redirection capability allowing data references in GUS displays to be redirected to alternative (HCI/OPC) data sources, thus allowing operational displays to be utilized “as is” in desktop platforms.

WebGUS ActiveX Control

The WebGUS product component that provides an ActiveX control that encapsulates a GUS run-time. This control can be embedded in ActiveX containers such as Internet Explorer web pages and GUS displays.

WebGUS Computer

A Desktop computer that has accessed the WebGUS Download and Configuration Web page at least once. The Web page access results in the download of the WebGUS components and the initialization of a Default Environment, visible as Registry entries in the HKEY_CURRENT_USER and HKEY_CLASSES_ROOT hives.

WebGUS Data Redirection

The WebGUS product components that provide re-direction capabilities for LCN data references in GUS displays. With this functionality, existing operational (control room) displays need not be altered in order to execute safely in a Desktop environment.

WebGUS Display

A GUS display rendered within a WebGUS control in runtime mode.

WebGUS Environment Definition

A named collection of settings that determine the macro behavior of WebGUS in a Desktop. An environment may be stored in a file and used to establish Registry settings. The environment defines:

- The set of OPC data access servers used by the client Desktop. The set includes information about each server: name, description, host, class ID and data collection rate.
- The set of timeout settings for the GUS Runtime and the HCI/OPC Data Access components.
- The set of mappings from the data sources used in an operational display to the OPC data access servers available at the client desktop.
- The setting for defining how to rename the point reference in the operational display to a convention understood by the OPC data access server.

WebGUS Environment Definition Configuration

The process of entering or modifying a WebGUS environment definition by using the WebGUS Environment Configuration pages distributed as part of the WebGUS product.

Web Site

A file system capable of holding HTML files.

Window

A rectangular area on a computer display in which application displays can be presented. In SafeView, the terms managed window and ViewPort are used interchangeably.

Window Category

A user-definable attribute that a workspace client can assign to its application window(s). This attribute provides a high degree of flexibility in managing such windows, because these windows can be differentiated by category and managed accordingly. Used by SafeView at runtime to associate a configured window specification with a real application window, at which time the application window is managed accordingly.

Window Group

For SafeView:

A collection of window specifications and/or other window groups within a workspace configuration. A workspace configuration is organized into a top-level window group and zero or more subgroups.

Window Management

The control of the display characteristics, (for example, position, style) of application windows. This is, essentially, workspace management.

W

Windows Management Instrumentation

Windows Management Instrumentation

Windows Management Instrumentation is a unified architecture for describing, accessing, and instrumenting objects. Part of this architecture is a large database of WMI classes used to carry out remote management tasks on specific objects.

Managed OU	An Organizational Unit is a Microsoft term for a container in the Windows 2000 Active Directory that stores users, computers, and other account objects. A TPS Domain or Console is derived from an OU. A Managed OU is a TPS Domain or Console that can be managed using the MMC Snap-in.
Monitored Domain	Monitored Domains automatically appear in the System Management Display. Monitored Domains may include a TPS Domain or Console, and non-TPS Domains
Managed Component	Managed components automatically appear on the System Management Display. Managed components include TPN Server, Fault Tolerant Ethernet (FTE), CL Server and NWDDDB Server.
Monitored Component	Monitored components are HCI or generic OPC servers, which are configured to be managed by the Redirection Manager (RDM).
CAS	Component Administration Service.

Windows Management Instrumentation Provider

A COM server that communicates with managed objects to access data and event notifications from a variety of sources.

Window of Output Focus

See Output Focus.

Window Specification

The basic unit within workspace configurations; contains a set of window-management properties such as position and size, etc., used to manage one or more application windows. These properties are applied to a real application window at runtime once SafeView determines that the window matches with the given window specification (or the window specification's group) based upon the match expression.

Window Title

For purposes of understanding SafeView, a window title is the text that appears in the title area of a top-level, application display window.

Workspace

SafeView

An operator's working user-interface environment. Microsoft Windows provides a basic workspace. The SafeView software extends the capabilities of that workspace.

TPS Builder

The Workspace is the main TPS Builder window, in which you work. You can open any number of project, library, and drawing windows in the workspace. You position these windows in the workspace in a way that allows you to access them as needed in an efficient manner.

Workspace Client

Any application that utilizes one or more SafeView API calls to register its display window(s) and uses other features described in the Application Program Interface section of this document. Any application display can be a workspace client by using the API `CreateWorkspaceClient`.

Workspace Configuration

A representation of a particular SafeView workspace, consisting of one or more window groups and window specifications. Created and edited by using the SafeView Text Editor and stored as a UNICODE text file.

W**Workspace Definition Language (WDL)**

Workspace Definition Language (WDL)

A language for the description of workspace window management.

Write

To save data into memory/disk—Store

To replace the value in a parameter with a new value.

Write Data

Values associated with the statement that were depopulated will be written to the external locations.

X

XML

Extensible Markup Language. A markup language that uses tags to make the data self-describing. Tags name the specific element of data that they delimit. For example;

```
<Server>  
    <ServerName>PHD Server 1</ServerName>  
    <Description>Server for Unit 1</Description>  
    <Host>\\Server001</Host>  
</Server>
```

X
XML

Z

Zener Diode

A silicon semiconductor device. It is frequently used as a voltage regulator to reduce the electrical energy entering over conductors into areas where explosive, hazardous atmospheres exist, thereby avoiding electrical ignition of the atmosphere.

Zip Drive

Takes a removable media Zip disk, which can hold 100 MB or 250 MB of data

Reader Comments

Fax Transmittal		Fax No.: (602) 313-4212
Reader Comments		
To:	Information Development	
From:	Name	Date:
	Title:	
	Company:	
	Address:	
	City:	State: Zip:
	Telephone:	Fax:
TPS Terms and Definitions , TP43-230 Release Release Independent, 12/03		
Comments: _____ _____ _____ _____ _____		
You may also call 800-343-0228 (available in the 48 contiguous states) or write to: Honeywell Industry Solutions 2500 West Union Hills Drive Phoenix, Arizona 85027		
Knowledge Building Tools		

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